

**VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY**



NGUYEN DUY CHIEN

**AUTOMATING QUESTION GENERATION
FROM TEXTBOOK CONTENT FOR SELF-STUDY**

BACHELOR THESIS

Major: Computer Science

Supervisor: Nguyen Thi Ngoc Diep, Ph.D

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Declaration

I declare that I am the sole author of this thesis and that it has not been previously submitted for any other academic or professional qualification. I confirm that the work submitted results from my efforts, supplemented by the valuable guidance and support of Nguyen Thi Ngoc Diep, Ph.D, for the main purpose of graduation.

All information, data, and results included in this thesis are original and have not been plagiarized from any other sources. All references used in this thesis are duly cited and listed in the reference section.

I take full responsibility for the content and integrity of this thesis, under the regulations of the VNU University of Engineering and Technology.

Ha Noi, 2025
Student

Nguyen Duy Chien

Supervisor's Approval

"I hereby approve that the thesis is accepted in its present form for committee evaluation as part of the requirements for the Bachelor's Degree in Computer Science at the University of Engineering and Technology."

Ha Noi, 2025
Supervisor

Nguyen Thi Ngoc Diep

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Abstract

In the era of advanced artificial intelligence, the demand for self-directed learning continues to rise. However, existing online learning frameworks often lack effective tools for practice and assessment, while content creation remains labor-intensive for educators. This highlights the need for automated, scalable solutions, particularly in automatic question generation, where traditional methods struggle with contextual accuracy in long documents.

This thesis introduces a Retrieval-Augmented Question-Answer Generation (RAGQA) pipeline to address these challenges and foster scalable automation in self-learning environments. Leveraging the strengths of RAG, RAGQA enables the dynamic generation of context-aware, pedagogically relevant, and personalized questions directly from educational materials in PDF format. This automated question generation framework supports more objective and adaptive self-assessment by aligning generated content closely with the source material.

RAGQA comprises three primary stages. First, input documents are segmented into semantically coherent chunks. These segments are then embedded and stored in a vector database. When a learner seeks to generate practice questions on a specific topic, a query retrieves the top-k most relevant document chunks based on vector similarity, which are then supplied as contextual input to a large language model. The model generates questions grounded in the retrieved content. To complete the learning loop, the system incorporates an evaluation component that assesses student responses using a large language model in conjunction with a carefully designed rubric.

Experimental results across diverse documents show that RAGQA consistently produces high-quality questions aligned with Bloom's taxonomy, with accuracy exceeding 90%. Moreover, other experiment results demonstrate its potential as a robust, intelligent tool for enhancing autonomous learning and formative assessment in modern education.

In conclusion, RAGQA presents a practical and scalable solution for automating the question generation process, thereby enhancing self-directed learning and formative assessment. By building on the strengths of RAG, the pipeline offers a foundational framework for future development of intelligent educational systems that can better support lifelong learning and the evolving demands of modern education.

Keywords: *Retrieval-Augmented Generation, Question Generation, Large Language Model, Prompt Engineering, Self-assessment.*

Tóm tắt

Trong kỷ nguyên của trí tuệ nhân tạo tiên tiến, nhu cầu về học tập cũng như tự học không ngừng tăng lên. Tuy nhiên, các khung học tập trực tuyến hiện tại thường thiếu các công cụ hiệu quả cho việc thực hành và đánh giá, trong khi việc tạo nội dung vẫn là một quá trình tốn nhiều công sức đối với các nhà giáo dục. Điều này làm nổi bật nhu cầu về các giải pháp tự động hóa và có khả năng mở rộng, đặc biệt là trong lĩnh vực tạo câu hỏi tự động, nơi mà các phương pháp truyền thống gặp khó khăn trong việc đảm bảo độ chính xác ngữ cảnh đối với các tài liệu dài.

Khóa luận này giới thiệu hệ thống Retrieval-Augmented Question-Answer Generation (RAGQA) để giải quyết những thách thức này và thúc đẩy tự động hóa trong tự học. Bằng cách tận dụng sức mạnh của RAG, RAGQA cho phép tạo ra các câu hỏi đa dạng, có nhận thức ngữ cảnh, phù hợp về mặt sư phạm và được cá nhân hóa trực tiếp từ các tài liệu giáo dục ở định dạng PDF. Khung tạo câu hỏi tự động này hỗ trợ việc tự đánh giá khách quan và thích ứng hơn bằng cách liên kết chặt chẽ nội dung được tạo ra với tài liệu nguồn.

RAGQA bao gồm ba phần chính, đầu tiên, các tài liệu đầu vào được chia thành các đoạn có ý nghĩa ngữ nghĩa. Các đoạn này sau đó được mã hóa và lưu trữ trong một cơ sở dữ liệu vector. Khi người học muốn tạo câu hỏi thực hành về một chủ đề cụ thể, truy xuất top-k đoạn tài liệu liên quan nhất dựa trên độ tương đồng vector, sau đó được sử dụng làm đầu vào ngữ cảnh cho một mô hình ngôn ngữ lớn. Mô hình này tạo ra các câu hỏi dựa trên nội dung được truy xuất. Để hoàn tất quá trình, hệ thống tích hợp một thành phần đánh giá sử dụng một mô hình ngôn ngữ lớn cùng với một bộ tiêu chí đánh giá được thiết kế cẩn thận để đánh giá câu trả lời của học sinh.

Kết quả thực nghiệm trên nhiều tài liệu đa dạng cho thấy RAGQA có khả năng tạo ra các câu hỏi chất lượng cao phù hợp với Bloom's taxonomy, với độ chính xác vượt quá 90% ở tất cả các mức độ. Hơn nữa, các kết quả thực nghiệm khác chứng minh tiềm năng của nó như một công cụ thông minh và mạnh mẽ để nâng cao việc học tập tự chủ và đánh giá hình thành trong giáo dục hiện đại.

Tóm lại, RAGQA đưa ra một giải pháp thực tiễn và có khả năng mở rộng cho việc tự động hóa quá trình tạo câu hỏi, từ đó nâng cao việc học tập tự định hướng và đánh giá hình thành. Bằng cách xây dựng trên nền tảng sức mạnh của RAG, hệ thống này cung cấp một khung cơ sở cho việc phát triển các hệ thống giáo dục thông minh trong tương lai, có thể hỗ trợ tốt hơn cho việc học tập suốt đời và đáp ứng các nhu cầu đang thay đổi của giáo dục hiện đại.

Từ khóa: *Tạo sinh tăng cường nhờ truy xuất, Tạo sinh câu hỏi, Mô hình ngôn ngữ lớn, Thiết kế đầu vào, Tự đánh giá*

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Acronyms

AQG Automating Question Generation

LLM Large Language Model

MCQ Multiple-Choice Question

NLP Natural Language Processing

Q&A Question & Answer

QG Question Generation

RAG Retrieval-Augmented Generation

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Chapter 1

Introduction

1.1 Motivation

1.1.1 The Demand for Studying and Lifelong Learning

In the modern world, education is no longer confined to early life or single institutions. Social, economic, and technological changes have shifted learning from a one-time experience to a continuous journey. On the social level, new career patterns and the demands of a globalized workforce have raised expectations for continual skill development. Many people today change jobs and careers multiple times throughout their lives, requiring them to learn new skills as industries evolve. For example, roles like data scientist or renewable energy technician did not exist a decade ago, illustrating how new professions demand new knowledge and abilities. At the same time, demographic shifts and longer life expectancy mean that people may spend several decades in the workforce, necessitating ongoing learning to keep their skills relevant.

Economic forces drive the push toward lifelong learning. We live in a knowledge-based economy where intellectual and technical skills are core assets. Advances in automation and artificial intelligence are transforming industries rapidly: routine tasks are increasingly handled by machines, while human work often shifts toward more complex, creative, and analytical tasks. As a result, specific skills can become outdated in a matter of years. For example, advances in software and machinery can make older tools and programming languages obsolete, prompting workers to seek retraining. Employers now emphasize continuous training and upskilling, and governments support adult education programs and workforce development initiatives. In many countries, national policies highlight lifelong learning as a priority for both economic growth and social well-being.

On the technological front, the proliferation of the internet and digital tools has made learning more accessible and necessary. In many fields, the body of knowledge doubles every few years or even months, so material learned in formal schooling can quickly become obsolete without ongoing refreshers. The internet offers an abun-

dance of learning resources: online courses, educational videos, interactive tutorials, and global knowledge networks. These platforms allow people to learn anytime and anywhere, breaking down geographical and scheduling barriers. For instance, someone learning a new language can practice daily using a mobile app, while an engineer can take an online tutorial to learn a new programming framework over a weekend. The ease of accessing up-to-date information means that learning can happen in small increments throughout life. Even after initial education, people can immediately apply and update what they learn by exploring new topics as they arise in their work and daily life.

A combination of social change, economic competition, and technological innovation has created a global demand for continuous education. Education is increasingly viewed as a lifelong process, not a fixed phase that ends at graduation. People now recognize that achieving career goals or keeping pace with a changing world requires regularly updating their knowledge and skills. Educational institutions, employers, and policymakers respond to this need by offering more flexible learning opportunities, such as micro-credentials, continuing education programs, evening classes, and online courses. However, each person's time is limited, and it is impossible to have enough classes and courses to meet the ever-increasing demand for learning. This creates an urgent need for an effective self-learning method.

1.1.2 Self-Study and Its Weaknesses

Self-study, sometimes called self-directed learning or autodidactic learning, refers to the process by which individuals take initiative in planning, executing, and evaluating their own learning activities without formal instruction. In the context of modern education and lifelong learning, self-study often involves using books, online courses, educational videos, and open educational resources independently. With the rise of the internet, people can access vast libraries of information and courses (Massive Open Online Courses, or MOOCs) at their own pace. This means a motivated learner can choose topics of interest and study from anywhere in the world without enrolling in a traditional program.

Self-study offers a range of key benefits that make it an increasingly attractive approach to learning. One of its most significant advantages is flexibility: learners can study at their own pace, on their schedule, and in environments that suit them best. They have the freedom to pause and replay tutorial videos, take breaks when needed, and resume learning without being bound to the pace of a formal class. This autonomy also enables greater personalization, as learners can focus on topics that are most relevant or interesting to them—for instance, someone interested in graphic design can delve into design tools and bypass unrelated material, customizing the learning path to fit personal goals. Additionally, self-study grants access to a diverse array of resources, including online lectures from international universities, interactive tutorials, and digital textbooks. Many of these materials are either free or low-cost, greatly expanding educational access on a global scale. Importantly, self-directed learning also fosters the

development of essential self-regulation skills such as time management, goal setting, and problem-solving, which are valuable in both personal and professional contexts.

However, self-study also comes with notable weaknesses and challenges that can hinder the learning process. One major issue is the lack of guidance and feedback; without teachers or mentors, learners may not realize when they misunderstand a concept, allowing misconceptions to persist uncorrected. For instance, a student misinterpreting a scientific principle might continue to build on that flawed understanding without timely intervention. Additionally, the absence of personalized feedback from assignments or exams can make it difficult to gauge progress. Motivation and discipline also pose significant hurdles—studying independently demands a high level of internal drive and consistency, which can be hard to sustain over time, especially without deadlines or social accountability. This often leads to low completion rates in self-paced online courses. Another common challenge is the lack of structure; with an overwhelming volume of resources available, learners may struggle to organize their studies effectively, skip essential topics, or jump between materials without a coherent plan, resulting in fragmented knowledge. Furthermore, self-study can be isolating, offering limited opportunities for real-time discussion or collaboration. The absence of peer interaction or instructor support can make learning feel disengaging and less dynamic. Finally, the quality and reliability of available materials vary greatly, and without expert guidance, learners may unknowingly rely on outdated or inaccurate sources, leading to confusion or wasted effort.

These weaknesses mean relying solely on self-study can be challenging. Learners often combine independent study with other methods to improve outcomes. For example, they might join study groups, use discussion forums to ask questions, or enroll in workshops and classes to get feedback and structure. Recognizing the limitations of pure self-study highlights the need for complementary tools and systems, such as automated quizzes, online Q&A, and mentorship programs, that provide guidance and structure for lifelong learners.

1.1.3 The Importance of Q&A in Learning and Self-Learning

Question-and-answer (Q&A) methods play a crucial role in learning and self-learning by actively engaging the learner in the process. Instead of passively reading or listening, the learner poses questions or answers questions about the material, which stimulates deeper cognitive processing. This can take many forms: a student might test themselves with flashcards, work through practice quizzes, or participate in discussion forums. In classrooms, teachers regularly ask questions to encourage students to recall information and think critically. In self-study, learners can use online quizzes or Q&A apps to create a similar interactive experience.

Q&A-based learning offers a powerful approach to education by leveraging active

engagement and instant feedback. It enhances memory retention through active recall, known as the testing effect, where retrieving information, like quizzing oneself on vocabulary, builds stronger memory connections than passive review. Students who test themselves on key concepts often retain them longer. The method also provides immediate feedback, allowing learners to spot and fix mistakes right away, much like an instructor correcting errors on the spot. This feedback loop, whether from automated quizzes or answer keys, ensures misunderstandings don't persist. Beyond retention, Q&A fosters deeper understanding and critical thinking by pushing learners to connect ideas actively—think of exploring "why" a principle applies rather than just memorizing it. The interactive, game-like nature of Q&A further boosts engagement and motivation, turning study into a rewarding challenge where each correct answer sparks a sense of achievement.

However, formulating questions during self-study remains a challenging and often overlooked aspect of the learning process. Despite being a crucial element for enhancing memory retention and achieving deeper understanding, there is still no optimal method for generating meaningful questions or objectively evaluating the answers. This limitation poses a significant barrier to independent learning. With the advancement of artificial intelligence, however, the efficiency of this process is improving. In particular, Retrieval-Augmented Generation (RAG) offers promising capabilities in this area. This very potential serves as the main motivation for this thesis, which aims to develop an effective question generation framework that closely aligns with source materials and supports learners in their self-study journey.

1.2 Automating Question Generation with RAGQA

1.2.1 Approach

Given the value of questions in learning, an important challenge is creating effective questions from educational content. Traditionally, instructors or textbook authors write exam questions or practice problems by hand. However, with the vast amount of content learners encounter (textbooks, articles, videos, etc.), manual question creation is laborious and often incomplete. This is where recent advances in artificial intelligence offer a promising solution: Retrieval-Augmented Generation (RAG) models.

Retrieval-Augmented Generation combines two steps: retrieving relevant information and then using a language model to generate text. In the context of question generation, a RAG model might first search a textbook chapter or lecture notes for key passages, and then use those passages as context to produce a natural-language question about them. The motivation for using RAG in question generation is to ensure that the AI-generated questions are both accurate and contextually grounded. Because the model retrieves specific information from the actual study material, the questions it creates are

guaranteed to relate directly to what the learner has read. This greatly reduces the chance of “hallucinated” questions that are irrelevant or incorrect. In other words, RAG helps keep the AI focused on the facts and concepts in the content.

Another advantage is variety and customization. A single textbook chapter can yield dozens of different questions, ranging from factual recall to analytical challenges. RAG models can generate this variety automatically, giving learners a broad set of practice questions. Moreover, because the retrieval step can be tailored to a user’s needs, the system can focus on specific subtopics or difficulty levels. For instance, a beginner studying cell biology might receive straightforward definition questions, while an advanced learner could get questions that ask for comparisons or applications of those concepts.

Using RAG for question generation significantly enhances the self-study process by delivering a tailored, efficient, and dynamic learning experience. RAG ensures contextual relevance by generating questions directly based on the material a learner is studying, making them focused and accurate. This is paired with its ability to provide a variety of questions, covering different aspects of the content, to offer a comprehensive set of practice problems. Additionally, the scalability of RAG models allows for the rapid creation of large question banks for any text or topic, bypassing the need for time-consuming manual question writing. As learners advance, RAG’s adaptivity shines, it fetches fresh context to produce updated questions, and can even target specific areas where more practice is needed. Together, these benefits make RAG an invaluable tool for self-study, blending personalization with efficiency.

1.2.2 Pipeline Components and Workflow

This thesis, therefore, proposes a Retrieval-Augmented Question Answer Generation (RAGQA) pipeline for automating question generation, inheriting from the general RAG pipeline. Its operational methodology is structured around three core stages: preprocessing, question generation, and evaluation, each meticulously crafted to contribute to a dynamic and responsive self-study environment. This tripartite architecture not only facilitates the generation of contextually pertinent practice questions but also delivers formative feedback, thereby fostering deeper learner engagement, promoting critical thinking, and enhancing knowledge retention.

RAGQA begins by systematically segmenting input documents into semantically coherent chunks. This segmentation process is foundational, as it ensures that the contextual integrity and nuanced meaning of the source material are preserved. Each chunk is then transformed into a high-dimensional vector embedding—a mathematical representation that encapsulates its semantic essence—using advanced embedding techniques. These embeddings are subsequently archived in a vector database, a storage mechanism optimized for efficient retrieval based on semantic similarity. The adoption

of vector embeddings is a critical design choice, as it enables the system to perform nuanced and precise matching of content with learner queries. This archival strategy underpins the system’s ability to rapidly access relevant information, laying a robust foundation for the subsequent stages of question generation and evaluation.

The flow continues by employing sophisticated similarity search algorithms to retrieve the top-k document segments that exhibit the highest semantic affinity to the learner’s query. This retrieval process is pivotal, as it ensures that the questions generated are not only aligned with the learner’s current focus but also deeply rooted in the original source material, thereby maximizing their educational relevance. The retrieved chunks are then utilized as contextual input for a large language model (LLM), a cutting-edge generative tool renowned for its ability to produce diverse and contextually rich outputs. Leveraging the LLM, RAGQA synthesizes questions that are both pertinent to the retrieved content and strategically designed to challenge and expand the learner’s comprehension. The use of an LLM in this capacity is particularly advantageous, as its generative capabilities allow for the creation of varied question types, spanning factual recall, analytical reasoning, and conceptual synthesis, catering to diverse learning styles and cognitive demands.

To complete the learning cycle, RAGQA incorporates an evaluation stage that rigorously assesses the learner’s responses. This module, which may utilize the same LLM or a separate one, is guided by a carefully designed rubric to ensure consistency, objectivity, and alignment with pedagogical best practices. The evaluation process goes beyond mere correctness, analyzing the depth and quality of the learner’s answers to provide a comprehensive assessment. The feedback generated is formative, offering actionable insights that reinforce the learner’s understanding, highlight strengths, and identify areas requiring further exploration. This stage is instrumental in closing the learning loop, as it transforms the act of answering questions into an iterative process of reflection and improvement. By delivering immediate and constructive feedback, RAGQA empowers learners to refine their knowledge and adopt a more targeted approach to their self-study efforts.

1.3 Objectives, Outline and Scope

The primary aim of this thesis is to assess the feasibility of employing the Retrieval-Augmented Generation (RAG) pipeline for the automation of question generation and to juxtapose its efficacy against conventional question generation techniques. The investigation is principally centered on the generation of questions derived from structured documents in PDF format, with a particular emphasis on textbooks, academic treatises, and peer-reviewed scientific literature. By leveraging the RAG framework, the study endeavors to architect a pipeline that generates questions that are both cogent and pertinent to the designated subject matter.

To systematically address the aforementioned objective, the thesis is delineated into five pivotal chapters:

- **Chapter 1: Introduction.** This inaugural chapter elucidates the intricacies of the question generation conundrum and underscores the escalating imperative for its automation. It concurrently delineates the constraints inherent in extant question generation methodologies that are tailored for self-directed learning paradigms. Furthermore, it proffers the RAG pipeline as an avant-garde solution for automating question generation, positing its potential to redress some of the enduring deficiencies in the domain.
- **Chapter 2: Related Works.** Commencing with a succinct exposition of germane domains, this chapter encapsulates the quintessential concepts underpinning Question Generation, Automated Question Generation, the RAG paradigm, and contemporaneous applications thereof. Through a meticulous synthesis of the extant corpus of literature, it aspires to furnish a holistic and profound comprehension of the problem landscape, the attendant impediments, and the lacunae that pervade the field.
- **Chapter 3: Methodology.** This chapter articulates the proposed methodological framework, which harnesses the RAG pipeline augmented with bespoke refinements to actualize the automation of question generation predicated on textbook content. It furnishes an exhaustive account of the modular construction and the stratagems employed to optimize the pipeline’s performance. Additionally, it deliberates on the arsenal of tools and frameworks requisitioned to effectuate the solution and to instantiate a demonstrative application.
- **Chapter 4: Experiments.** This segment delineates a series of meticulously crafted experiments tailored to interrogate the problem at hand. It encompasses the experimental architecture, the empirical outcomes garnered, and the inferences extrapolated from these findings, thereby proffering a rigorous appraisal of the proposed methodology.
- **Chapter 5: Conclusion.** The culminating chapter synthesizes the salient findings of the thesis, encapsulating its scholarly contributions and the broader ramifications of the research endeavor. It critically appraises the merits and demerits of the study and prognosticates prospective trajectories for ensuing research and development initiatives.

The scope of this thesis is primarily centered on the generation of questions derived from formatted documents, specifically PDFs, with a particular emphasis on books, textbooks, and published scientific research. Employing the Retrieval-Augmented Generation (RAG) framework, the study aims to architect an efficient workflow that yields questions that are both cogent and pertinent to the selected subject matter.

Chapter 2

Related Works

2.1 Traditional Question Generation Methods

Traditional question generation methods, developed primarily before the advent of modern machine learning and deep learning techniques, rely heavily on rule-based, template-based, or heuristic-driven approaches to transform textual input into questions. These methods leverage linguistic rules, syntactic transformations, and predefined patterns to generate questions, often with a focus on educational applications or natural language processing research. While foundational to the field of automated question generation, these approaches are characterized by their dependence on manual effort and domain-specific expertise, which limits their scalability and adaptability. This section provides an in-depth review of key works in traditional question generation, introduces additional relevant studies, and traces the evolution of these methods over time, highlighting their contributions and shortcomings.

One of the earliest notable efforts in traditional question generation is the work of Ruslan Mitkov and Le An Ha [24], who explored computer-aided generation of multiple-choice tests. Their approach used basic natural language processing techniques, such as keyword extraction and template matching, to identify important text terms and phrases and generate questions with plausible distractors. While this method was straightforward and computationally efficient, it produced questions that were often superficial, focusing on factual recall rather than deeper comprehension, due to its reliance on simple pattern-matching techniques.

Building on this foundation, Mitkov advanced the field with their "Computer-Aided Environment for Generating Multiple-Choice Test Items." [23] This system employed more sophisticated NLP techniques, including syntactic analysis and semantic similarity measures, to generate multiple-choice questions (MCQs). The process involved extracting key concepts from the text, identifying related terms to serve as distractors, and framing questions that test specific knowledge points. This approach was particularly

valuable in educational settings, where MCQs are widely used to assess student understanding. However, it required significant manual tuning to ensure the distractors were both plausible and distinct, highlighting a key limitation of traditional methods: their dependence on human intervention.

Another landmark contribution came from Heilman and Smith [11], who introduced the overgenerate-and-rank paradigm. Their method begins by applying syntactic transformation rules to an input sentence to produce a large pool of candidate questions. This approach was groundbreaking because it moved beyond rigid templates, enabling the generation of a wider variety of questions. However, its effectiveness depended on the quality of the transformation rules and the ranking model, both of which required extensive linguistic expertise to develop and refine.

Researchers further expanded the scope of traditional QG by focusing on questions that promote deeper learning [32]. Their work was part of the inaugural Question Generation Shared Task Evaluation Challenge, an initiative to standardize QG system evaluation. They emphasized generating questions that target higher levels of Bloom’s taxonomy, rather than simple recall. Their system combined rule-based transformations with heuristic scoring to prioritize pedagogically valuable questions. While influential in establishing benchmarks for QG research, their approach still relied on manually crafted rules, limiting its ability to scale across diverse texts or domains.

Beyond these foundational works, several other studies have enriched the landscape of traditional QG methods. Mazidi and Nielsen (2014) proposed a hybrid approach that integrates rule-based techniques with rudimentary machine learning [22]. This method improved upon purely rule-based systems by incorporating statistical techniques to prioritize sentences, but it still required manual rule design and struggled with generating questions that demand inferential reasoning. Labutov (2015) took a different tack with their template-based method tailored for scientific texts [9]. Their system uses a library of predefined question templates and populates them with entities and relationships extracted from the text using NLP tools like named entity recognition and dependency parsing. This approach excelled in domain-specific contexts, such as science education, but its reliance on a fixed set of templates restricted its flexibility. Adapting the system to new domains necessitated creating additional templates, a time-consuming process. Another relevant contribution is from Yao (2013), who explored semantics-based QG [40]. Their system uses semantic role labeling (SRL) to identify the roles of entities in a sentence (e.g., agent, action, object) and generate questions targeting these roles. This method marked a step forward in capturing semantic relationships, but its reliance on accurate SRL and predefined question patterns limited its robustness across varied texts.

Traditional QG methods have notable strengths. Their rule-based design ensures predictable, interpretable outputs, ideal for controlled settings like educational assessments. They generate reliable MCQs for specific concepts or excel in scientific domains

using templates. Unlike data-driven approaches, these methods require minimal datasets and computational resources, making them suitable for early research or resource-limited environments.

However, their drawbacks are significant. Handcrafted rules and templates limit scalability and adaptability. For instance, an overgenerate-and-rank approach requires domain-specific transformation rules, hindering generalization. Template-based systems struggle with texts outside their predefined scope. Additionally, these methods often lack the semantic depth for inferential or nuanced questions, producing grammatically correct but pedagogically shallow outputs, as noted in a 2020 systematic review. This has driven the shift toward automated, data-driven techniques.

In summary, traditional QG methods laid critical groundwork, offering practical tools and insights for automated question generation. They established key paradigms overgeneration, rule-based processing, and educational focus that continue to shape QG research. Later advancements refined these approaches, integrating sophisticated NLP techniques to enhance question quality. Over time, the field progressed from basic template matching to advanced syntactic and semantic methods, reflecting NLP’s evolution.

Still, the labor-intensive nature of these methods and their struggles with complex, diverse, or abstract content have fueled the adoption of modern approaches. Machine learning and deep learning, leveraging large datasets and neural architectures, generate questions with greater flexibility and semantic richness. These advancements, explored later in this thesis, build on traditional QG’s legacy while overcoming its core limitations.

2.2 Automating Question Generation

Automating Question Generation (AQG) represents a transformative paradigm in educational technology, aiming to alleviate the cognitive and temporal burdens associated with manual question crafting. AQG leverages computational techniques, particularly natural language processing (NLP) and machine learning, to produce questions from textual corpora, thereby facilitating scalable and adaptive learning environments. This field has garnered substantial attention due to its potential to enhance pedagogical outcomes across diverse educational contexts, including classroom instruction, self-directed learning, and professional training. By generating questions that probe comprehension, critical thinking, and application, AQG systems support formative assessment and foster deeper engagement with content [17].

Historically, question generation was a labor-intensive process reliant on human expertise, limiting its scalability. The advent of NLP technologies, such as syntactic parsing and semantic analysis, has enabled the development of systems capable of extracting key information from texts and formulating relevant questions [12]. Recent advancements in deep learning, particularly transformer-based models like BERT and

GPT, have further revolutionized AQG by enabling the generation of contextually nuanced and diverse questions [7, 29]. These models excel at understanding complex textual structures, making them well-suited for educational applications. AQG systems are now deployed in various domains, from generating factual recall questions to crafting higher-order questions that require analytical reasoning [3].

The significance of AQG is underscored by its ability to address the escalating demand for personalized and adaptive learning systems, especially with the proliferation of MOOCs and e-learning platforms. Research suggests that AQG can reduce the cost and effort of question construction, enabling adaptive testing, drill and practice exercises, and controlling question characteristics like difficulty and cognitive level [2]. Beyond education, AQG finds applications in validating knowledge bases, developing conversational agents, and enhancing question answering or machine reading comprehension systems for training and testing [36].

Contemporary AQG methodologies are predominantly grounded in advanced NLP frameworks, with deep learning models at their core. Transformer architectures, such as BERT [7] and T5 [30] has become the de facto standard due to its ability to capture bidirectional contextual relationships and generate coherent text. These models are typically fine-tuned on question-answering datasets, such as SQuAD [31], to adapt their language generation capabilities for question formulation. The process generally involves three stages: content selection, question formulation, and question ranking or filtering.

In content selection, AQG systems identify salient information from input texts using techniques like named entity recognition (NER) and keyphrase extraction [25]. For instance, a rule-based approach was proposed a rule-based approach to select answerable segments from texts, which serves as a foundation for many modern systems. Question formulation leverages sequence-to-sequence models or template-based methods, with recent works favoring transformer-based models for their superior contextual understanding. An attention-based sequence learning model for question generation was introduced, demonstrating significant improvements over rule-based systems by training end-to-end on the SQuAD dataset [8].

Transformer-based models, such as T5, frame question generation as a text-to-text task, where the input is a passage and an answer, and the output is the corresponding question. This approach was exemplified, conditions generation on both passage and answer, enhancing relevance and fluency [5]. Moreover, research has explored generating higher-order questions requiring critical thinking, moving beyond factual recall. A paragraph-level neural question generation model with maxout pointer and gated self-attention networks was proposed, achieving improved performance on complex question types [41]. The shift from traditional rule-based and template-based methods, which often struggled with scalability and diversity, to data-driven deep learning approaches, particularly transformers, underscores the current state-of-the-art in AQG methodolo-

gies.

Despite significant advancements, AQG faces several challenges that impede its widespread adoption in educational settings. One primary challenge is ensuring the diversity and quality of generated questions. Deep learning models, while capable of producing fluent questions, often lack variety, potentially generating similar questions for different inputs, which can reduce assessment effectiveness [17]. Another significant hurdle is controlling the difficulty level of generated questions, crucial for adaptive learning environments. Most systems lack mechanisms to adjust difficulty, limiting their applicability to diverse learner proficiencies [2].

Evaluation poses another challenge, with automatic metrics like BLEU and ROUGE often failing to capture pedagogical value, necessitating resource-intensive human evaluations [26]. Generating questions requiring critical thinking and reasoning, rather than factual recall, remains difficult, with current systems better suited for lower-order questions [41]. Additionally, ungrammaticality and nonsensical outputs, particularly in syntactic transformation approaches, are notable issues, with reporting 36.7% and 39.5% ungrammatical and nonsensical questions, respectively [13].

Opportunities for AQG are substantial, particularly in transforming educational practices. It can support personalized learning by generating questions tailored to individual needs and progress, integrating into intelligent tutoring systems for real-time feedback and assessment [18]. AQG can assist educators in creating teaching materials efficiently, reducing administrative burdens, and enabling focus on teaching. It also holds promise for self-study, providing learners with questions and feedback to test understanding independently, potentially enhancing e-learning platforms [36]. Ongoing research aims to address these challenges, leveraging advancements in NLP and machine learning to make AQG more effective and widely applicable.

2.2.1 Automating Question Generation from Textbook

Textbooks, as structured repositories of knowledge, contain critical concepts, definitions, and explanations, making them ideal for AQG. The process involves selecting relevant content and transforming it into questions that align with educational objectives, such as assessing comprehension or encouraging critical thinking. AQG from textbooks reduces the temporal and cognitive burdens on educators, enabling the creation of diverse assessment materials efficiently. However, identifying question-worthy content and ensuring pedagogical relevance remain significant challenges.

Content selection is a pivotal step in AQG, requiring the identification of salient information within textbook passages. Advanced methodologies employ pedagogically-driven mechanisms, utilizing linguistic features and machine learning to prioritize sentences encapsulating definitions and key facts [36]. Techniques such as named entity recognition (NER) and keyphrase extraction enhance precision in extracting relevant

content [25]. Rule-based approaches have historically used gap selection and text summarization, often leveraging external resources like Wikipedia, with systems incorporating constituency parsers and semantic role labelers to select answerable segments, achieving up to 71% acceptable question rates [12]. Neural network-based methods, including encoder-decoder architectures with attention mechanisms, have further improved selection accuracy, trained on datasets like SQuAD, yielding BLEU scores of 12.28 and ROUGE scores of 39.75 [8].

Once content is selected, question formulation transforms it into interrogative forms. Traditional template-based methods fill predefined patterns with extracted keywords, though they often lack flexibility [12]. Data-driven approaches, particularly sequence-to-sequence models, have surpassed these limitations. An attention-based model trained end-to-end on SQuAD significantly outperformed rule-based systems, leveraging encoder-decoder architectures to focus on relevant text segments [8]. Transformer-based models, such as T5, fine-tuned on question-answering datasets, excel in generating contextually relevant questions, achieving BLEU scores of 25.42 and ROUGE scores of 45.75 [30]. Reinforcement learning has been integrated to optimize question quality, aligning outputs with pedagogical goals [10]. Domain-specific applications, like generating gap-fill questions from biological textbooks, demonstrate AQG’s adaptability [1]. For multiple-choice questions (MCQs), generating effective distractors is crucial, requiring an understanding of common misconceptions. Semantic similarity measures and domain-specific knowledge bases craft challenging distractors, enhancing question educational value [21]. These distractors ensure MCQs test nuanced understanding, not merely recall.

Evaluation combines automatic metrics (BLEU, METEOR, ROUGE) and human judgment to assess grammaticality, relevance, and pedagogical soundness. Studies indicate that AQG-generated questions, particularly in biology, are comparable to human-expert quality [17]. Case studies highlight AQG’s potential to reduce assessment creation time, offering scalable solutions for educational resource development [36].

2.2.2 Automating Question Generation for Self-Study

AQG for self-study empowers learners to test their knowledge independently, providing questions with feedback to facilitate learning. Immediate feedback identifies weaknesses and reinforces understanding, crucial for online courses and digital textbooks. AQG systems generate diverse question types and explanations, supporting personalized learning experiences.

A prominent approach generates MCQs with distractors and explanations for each option’s correctness, enhancing learner understanding through detailed feedback [21]. Semantic similarity and domain knowledge ensure that distractors challenge misconceptions [21]. Open-ended questions with model answers allow learners to compare re-

sponses, with explanations often extracted from the source text for contextual relevance. Short-answer questions, fill-in-the-blank, and essay prompts cater to varied cognitive demands, encouraging critical thinking and synthesis.

Feedback is critical for self-study, requiring informative explanations of answer correctness. However, generating high-quality feedback is challenging, with many systems offering basic prompts like retry suggestions [2]. Advanced systems use natural language generation to produce custom explanations, leveraging source text or knowledge bases [21]. Template-based feedback fills specific details, while adaptive systems track learner progress to provide personalized responses.

A learner-centric study with college students demonstrated AQG’s efficacy in improving reading comprehension, with a medium effect size (Cohen’s $d = 0.63$) [35]. Participants benefited from generated questions, even when identified as computer-generated, with no significant negative effects on unrelated outcomes [35]. Automatic metrics assess question quality, but human evaluations ensure pedagogical value, highlighting the need for improved feedback depth [2].

AQG offers substantial opportunities to transform education. For textbooks, it supports scalable assessment creation, integrating into intelligent tutoring systems for real-time feedback [36]. For self-study, AQG enhances e-learning platforms, providing tailored questions and feedback [35]. Future research aims to improve question diversity, difficulty control, and feedback quality using advanced NLP models. Integrating multimodal inputs (e.g., images) and developing standardized evaluation metrics could further enhance AQG’s applicability.

2.3 Retrieval-Augmented Question Generation

2.3.1 Retrieval-Augmented Generation (RAG)

Large Language Models (LLMs), such as GPT-3 [4], BERT [7], LLaMA [37], DeepSeek [6], and Grok3, etc, have fundamentally transformed the discipline of natural language processing (NLP) by facilitating advanced text comprehension and generation capabilities. These models, predominantly constructed upon the transformer architecture [38], employ self-attention mechanisms to discern long-range dependencies within textual data. Typically, LLMs undergo pre-training on expansive text corpora using unsupervised learning objectives, such as next-word prediction or masked language modeling, followed by fine-tuning on specific tasks, thereby achieving exemplary performance in domains like question answering, text completion, and dialogue generation [29]. Their efficacy stems from their capacity to encapsulate extensive linguistic and semantic knowledge within their parameters, yielding outputs that emulate human-like fluency. Nevertheless, LLMs encounter substantial limitations in knowledge-intensive tasks necessitating current or domain-specific information. Given their reliance on static

training datasets, their knowledge remains fixed, rendering them incapable of accessing real-time data or specialized insights absent from their initial corpora [28]. This constraint often results in outdated or erroneous responses, compounded by "hallucinations," wherein models produce plausible yet factually inaccurate content due to an inability to validate or refresh their knowledge base [15]. Such deficiencies underscore the imperative for methodologies that augment LLMs with dynamic external knowledge sources to bolster their efficacy in tasks demanding precision and contextual relevance.

To mitigate these shortcomings, Retrieval-Augmented Generation (RAG) was introduced in 2021 [20]. RAG constitutes a hybrid paradigm that synergistically integrates retrieval mechanisms with generative models, empowering the system to consult external knowledge repositories during text synthesis. This approach amalgamates parametric knowledge—embedded within the model’s parameters—with non-parametric knowledge sourced from external documents, thereby enhancing the generation of accurate and contextually pertinent outputs. The original pipeline is demonstrated in Figure 2.1.

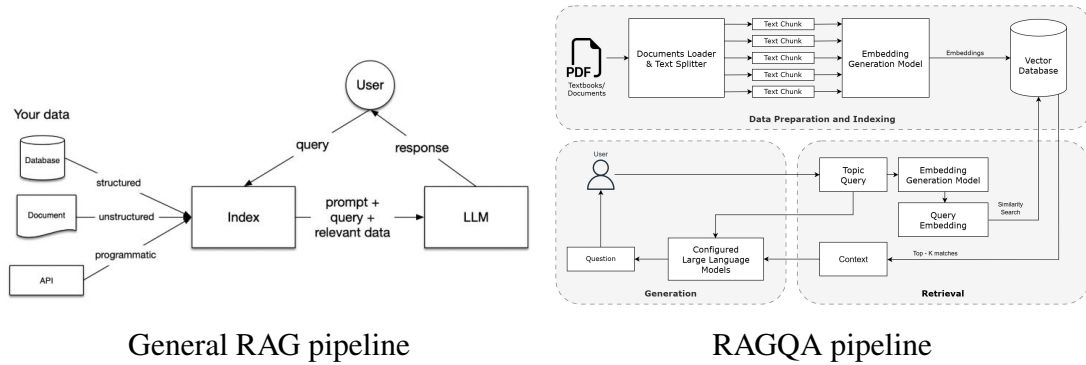


Figure 2.1: *Correlation between the general pipeline and the proposed pipeline. The proposed pipeline introduces several notable differences. It incorporates embedding models to enhance both the document chunking process and the retrieval mechanism. In contrast, general pipelines typically focus more on answering questions rather than generating them, which leads to some structural and functional differences between the two approaches.*

Operationally, RAG employs a retriever, often leveraging dense vector representations as exemplified by Dense Passage Retrieval (DPR) [16], to pinpoint relevant documents within a vast corpus. Subsequently, a generator, typically a sequence-to-sequence model such as BART [19] or T5 [30], synthesizes text informed by these retrieved documents, grounding the output in verifiable external data. Since its inception, RAG has spurred a plethora of refinements, including optimizations in retrieval through relevance feedback and advanced ranking algorithms [14], as well as integrations with structured knowledge frameworks like knowledge graphs [39]. Empirical evaluations have substantiated RAG’s versatility across diverse NLP applications, affirming its capacity to enrich model outputs with external context [34].

2.3.2 RAG for Automating Question Generation

The automation of question generation represents a pivotal endeavor across multiple domains, including educational technology, content creation, and interactive systems such as chatbots. This process entails crafting questions from given contexts or texts to assess comprehension, foster learning, or stimulate user engagement. The emergence of LLMs has revolutionized this landscape, enabling the production of diverse, contextually apt questions via sequence-to-sequence architectures that capitalize on their linguistic proficiency [8]. However, LLMs alone may falter in ensuring factual accuracy or relevance when requisite information transcends the input text or their pre-trained knowledge. RAG emerges as a potent enhancer in this context, supplying additional context through retrieved documents and facilitating the formulation of questions that extend beyond immediate inputs. This attribute is invaluable for crafting queries that probe deeper comprehension or require external knowledge, as required in educational assessments or knowledge-driven quizzes.

One notable framework that enhances question generation by selecting relevant sentences from a given context has been proposed by researchers [27]. Their model employs a selector module to identify key sentences and a generator module to produce questions based on this distilled information. This selector-generator framework implicitly incorporates retrieval-like mechanisms, as the selection process mirrors the retrieval of pertinent information to inform question generation. Their approach has shown improved performance on benchmark datasets like SQuAD, highlighting the potential of retrieval-enhanced methods in producing high-quality questions.

Other related work includes efforts to use question generation to improve retrieval in question answering systems. For instance, researchers explored zero-shot question generation to re-rank retrieved passages, demonstrating how generated questions can enhance the relevance of retrieved content [33]. While their focus is on question answering, the use of question generation in a retrieval context suggests potential applications for RAG in question generation tasks.

The integration of RAG into question generation offers several advantages. Firstly, RAG enables access to a broader knowledge base, allowing the generation of questions that incorporate information beyond the immediate input text. This is particularly beneficial for creating questions that test deeper understanding or require external knowledge, such as in educational assessments or knowledge-based quizzes. Secondly, by grounding the generation process in retrieved documents, RAG mitigates the risk of hallucinations, incorrect or fabricated information generated by LLMs, resulting in more reliable and factually accurate questions. This is critical in applications where accuracy is paramount, such as academic testing.

However, RAG-based question generation also faces challenges. The effectiveness of the approach heavily depends on the quality of the retrieval step. If the retriever fails

to identify relevant or accurate documents, the generated questions may be misleading or irrelevant. For example, irrelevant retrieved content could lead to questions that are off-topic or factually incorrect. Additionally, the computational overhead of combining retrieval and generation processes can be significant, posing scalability issues for large-scale applications or resource-constrained environments. The complexity of integrating and fine-tuning the retriever and generator components further complicates deployment.

Another consideration is the limited number of academic studies specifically addressing RAG for question generation. While projects and frameworks demonstrate potential, the field is still emerging, and more research is needed to establish best practices and evaluate performance across diverse datasets and applications.

Retrieval-Augmented Generation represents a promising direction for automating question generation, offering a robust framework that combines the strengths of retrieval and generation. By leveraging external knowledge sources, RAG-based systems can produce high-quality, contextually informed questions that enhance educational tools, content creation, and interactive systems. Despite challenges such as retrieval accuracy and computational demands, the ability of RAG to reduce hallucinations and incorporate dynamic knowledge makes it a valuable tool for advancing question generation. As research in this area progresses, RAG is likely to play an increasingly significant role in developing sophisticated question generation systems.

2.4 Recent Applications in Automating Question Generation

The practical implementation of Question Generation has led to the development of several AI-powered tools that enhance educational processes. These tools leverage advanced models to generate questions, provide feedback, and support personalized learning, significantly reducing the workload for educators. Several popular tools are systematically compared and enumerated in the table below.

Some specialized tools have emerged to support educators in creating assessments. QuestionWell, developed with input from educators, generates standards-aligned questions from text inputs, reducing the time required to create quizzes and worksheets. Quillionz, described as the world's first AI-powered question generator, excels at producing questions from structured, factual content, supporting various question types like multiple-choice and "Wh" questions. QuizGecko offers a user-friendly platform for generating quizzes from existing content, with features like auto-grading and customizable settings. Other tools, such as EdutorAI and Quizbot, further expand capabilities by supporting inputs like PDFs, videos, and web links, catering to diverse educational needs.

While these tools offer significant benefits, they also raise ethical considerations. Ensuring the accuracy and fairness of generated questions is paramount, as biases in AI

Tool	Usage	Advantages	Disadvantages
Quizlet Q-Chat	• AI chat tutor in Quizlet • Flashcard-based quizzes • Beta (2023, OpenAI API)	• Adapts to learning pace • Dynamic explanation rephrasing • Instant answer correction	• Quizlet platform lock-in • Premium subscription needed • Beta feature limitations
Socratic by Google	• Photo-based homework help • Step-by-step explanations • 2023 AI relaunch	• Multi-subject support • Follow-up Q&A chat • Camera input flexibility	• High-school level focus • Math symbol recognition issues • Risk of student over-reliance
LMS Education Copilots	• AI in Canvas/Blackboard • Course Q&A automation • Syllabus query support	• 24/7 availability • Reduces instructor workload • Institutional data integration	• Generic automated responses • Privacy concerns • LMS platform dependency
Khan Academy AI Tutor	• Personalized learning paths • Interactive exercises • Free access platform	• Progress tracking • Adaptive difficulty • No subscription required	• STEM-focused content • Limited conversation ability • Internet dependency
Quizizz AI	• Auto-generated quizzes • Content-to-quiz conversion • Teacher-focused tool	• Time-saving batch creation • Gamified learning • Basic analytics included	• Input quality dependency • Limited question types • Advanced features paywalled

Table 2.1: *Comparison of recent Automated Question Generation tools. Many automatic question generation tools are now available, offering significant benefits. However, these tools often come with certain dependencies on specific platforms, which can limit their accessibility and usability for a broader audience.*

models can lead to inequitable assessments. Privacy concerns arise when student data is used to train or personalize systems, necessitating robust data protection measures. Additionally, the reliance on AI-generated content requires instructors to review outputs for accuracy and relevance, maintaining human oversight in the educational process.

The future of Automating Question Generation lies in addressing current limitations and exploring new possibilities. Research is likely to focus on generating questions from multimodal content, such as videos and interactive simulations, to support diverse learning environments. Incorporating real-time student feedback into these systems could further personalize question generation, adapting to individual learning needs. Advances in evaluation metrics and ethical AI frameworks will also be critical to ensuring the reliability and fairness of these tools.

Chapter 3

Methodology & Implementation

3.1 Proposed pipeline for Automating Question Generation from Textbook

The proposed pipeline for automated question generation draws inspiration from Retrieval-Augmented Generation (RAG) frameworks but reorients the workflow toward synthesizing contextually relevant questions rather than answers. As shown in Figure 3.1, this structured approach comprises three interconnected stages: document processing, data preparation and indexing, and generation. Each phase is designed to systematically transform raw textual inputs into pedagogically meaningful questions while addressing computational and pedagogical constraints.

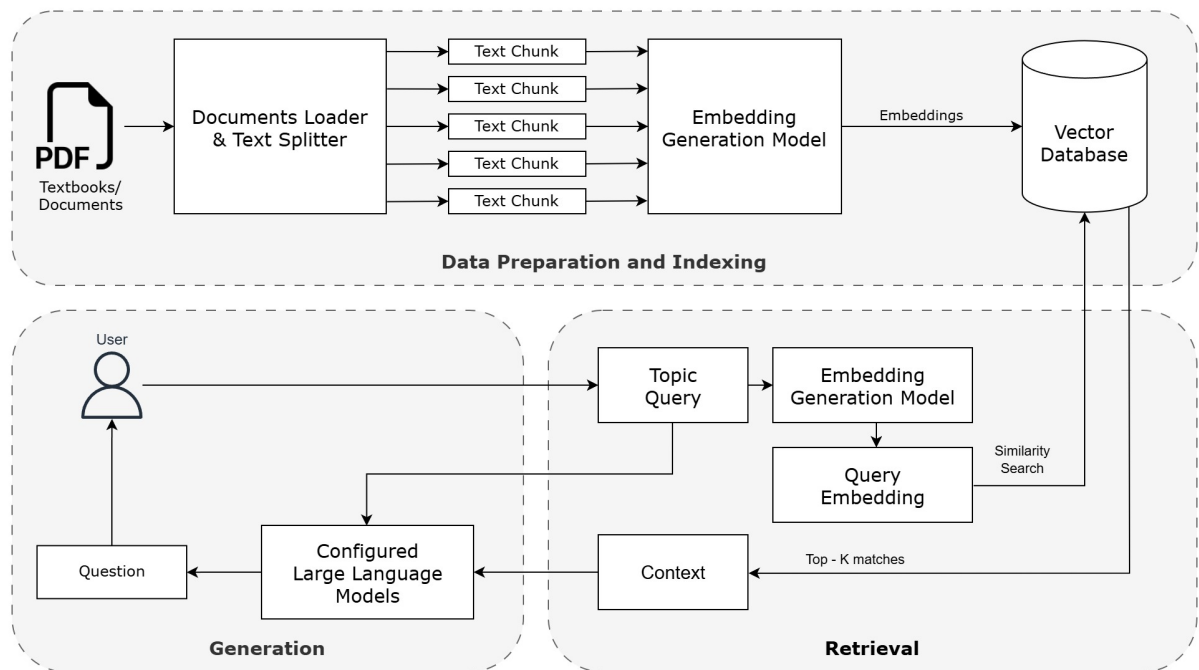


Figure 3.1: *Proposed pipeline for Automating Question Generation from Textbook*

The data preparation and indexing phase focuses on ingesting and preparing source materials such as textbooks or PDFs. Documents are first segmented into smaller text chunks to accommodate the token limitations of language models and preserve semantic coherence. The chunking strategy, whether fixed-size windows or semantic segmentation, critically influences downstream performance, as overly fragmented text risks loss of context, while excessively large segments may introduce noise. Subsequently, an embedding generation model converts these chunks into high-dimensional vector representations. These embeddings are indexed into a vector database to enable efficient similarity searches. The choice of embedding model and database architecture directly impacts retrieval accuracy and scalability, particularly for large-scale educational datasets.

In the retrieval phase, user-provided topic queries are processed to guide question generation. The query is embedded using the same model employed for document processing, ensuring compatibility in the vector space. A similarity search algorithm retrieves the top-K most relevant text chunks from the database, balancing precision and breadth through adjustable parameters. Combined with the original query, retrieved chunks form a contextual prompt for configured large language models. Key considerations here include mitigating retrieval bias by diversifying source content and dynamically adjusting the value of K based on query complexity. For instance, intricate topics may benefit from a higher K to capture broader context, while straightforward queries might require fewer, highly focused segments.

The generation phase synthesizes questions by leveraging the retrieved context and the generative capabilities of LLMs. Prompt engineering is pivotal in directing models to produce pedagogically appropriate outputs. For example, prompts may specify question types, cognitive levels, or difficulty tiers. Post-processing steps, such as filtering redundant questions or validating answer ability, further refine outputs. Challenges in this stage include minimizing LLM hallucinations, ensuring questions adhere strictly to retrieved evidence, and maintaining diversity in question formats to cater to varied educational objectives.

3.2 Data Preparation and Indexing

3.2.1 Documents Chunking

Document chunking serves as a critical preprocessing step in Retrieval-Augmented Generation (RAG) systems, enabling the transformation of lengthy, unstructured textual data into manageable segments that align with the operational constraints of modern language models. Language models such as BERT and GPT3 operate within fixed token windows, typically ranging from 512 to 4,096 tokens, necessitating the division of extensive documents like textbooks or research papers into smaller units [4, 7]. This segmentation ensures that downstream tasks, such as semantic retrieval and context-aware

generation, can function efficiently. However, the process is not merely technical; it also demands careful consideration of semantic coherence. Poorly executed chunking risks fragmenting key concepts, isolating cross-references, or diluting the pedagogical intent of educational materials, thereby undermining the system’s ability to generate accurate or contextually relevant outputs.

The challenge intensifies when processing long-form documents such as textbooks, which often feature hierarchical structures, embedded diagrams, and dense interconnections between concepts. For instance, a biology textbook might intersperse explanations of cellular processes with diagrams of mitosis, requiring chunking strategies that preserve both textual and visual context.

Traditional methods like fixed-size chunking, while computationally straightforward, often fail to account for these nuances, leading to disjointed segments that disrupt the logical flow of information. As RAG systems increasingly find applications in educational technology and automated content curation, optimizing chunking techniques for such documents has become a focal area of research. Two of the chunking algorithms were experimented with in the thesis is described as below:

Algorithm 1 Fixed-Size Sliding Window Chunking

Input: Document D ,
max tokens per
chunk $L = 256$,
overlap $O = 64$

Output: Chunks $\mathcal{C}$

```

1: Tokenize  $D$  into tokens  $T = (t_1, \dots, t_N)$ 
2:  $\mathcal{C} \leftarrow []$ ,  $s \leftarrow 1$ 
3: while  $s \leq N$  do
4:    $e \leftarrow \min(s + L - 1, N)$ 
5:    $C \leftarrow (t_s, \dots, t_e)$ 
6:   append  $C$  to  $\mathcal{C}$ 
7:   if  $e = N$  then
     break
8:   end if
9:    $s \leftarrow e - O + 1$ 
10: end while
11: return  $\mathcal{C}$ 

```

Algorithm 2 Semantic Chunking via Embedding Clustering

Input: Document D , sentence splitter, embedding model $\phi(\cdot)$, similarity threshold τ

Output: Chunks $\mathcal{S}$

```

1: Split  $D$  into sentences  $S = (s_1, \dots, s_M)$ 
2: Compute embeddings  $E_i = \phi(s_i)$  for  $i = 1, \dots, M$ 
3: Initialize  $\mathcal{S} \leftarrow []$ , current chunk  $C \leftarrow []$ , centroid  $c \leftarrow \mathbf{0}$ 
4: for  $i = 1$  to  $M$  do
5:    $e \leftarrow E_i$ 
6:   if  $\|e - c\|_2 \leq \tau$  or  $C = []$  then
7:     append  $s_i$  to  $C$ 
8:     update centroid  $c \leftarrow \frac{|C|-1}{|C|} c + \frac{1}{|C|} e$ 
9:   else
10:    append  $C$  to  $\mathcal{S}$ 
11:     $C \leftarrow [s_i]$ ,  $c \leftarrow e$ 
12:  end if
13: end for
14: if  $C \neq []$  then append  $C$  to  $\mathcal{S}$ 
15: end if
16: return  $\mathcal{S}$ 

```

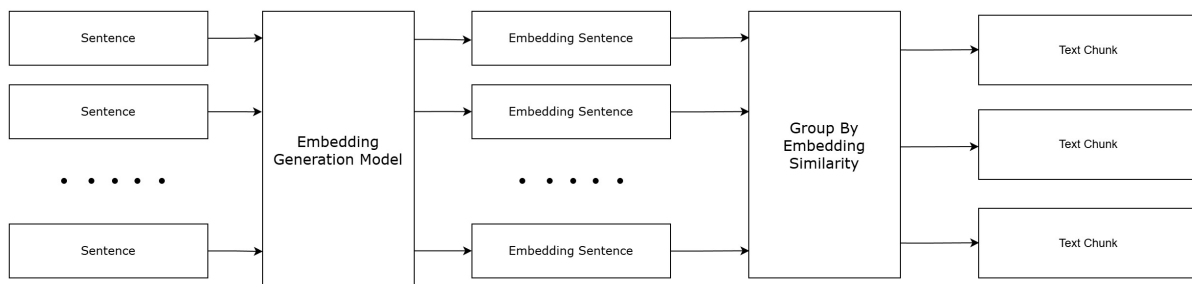


Figure 3.2: Semantic Chunking workflow

This thesis employs the sophisticated technique of semantic chunking as demonstrated in Figure 3.2, whereby a document is systematically segmented into discrete sentences utilizing a text splitter. Subsequently, these sentences are subjected to embedding processes, transforming them into vector representations. These embeddings are then clustered into cohesive chunks based on their semantic similarity, as determined by their embedding proximity. This methodological approach enhances the precision of information retrieval and processing by ensuring that the resulting chunks are semantically coherent, thereby facilitating more nuanced and contextually relevant analyses. The rationale for adopting this technique lies in its capacity to preserve the intrinsic meaning of the text while enabling efficient data handling and improved interpretability in downstream applications.

3.2.2 Documents Embedding

Embeddings are a way to represent words, sentences, or even entire documents as dense vectors in a high-dimensional space. The purpose of embeddings is to capture the semantic meaning of text, such that words or phrases with similar meanings are located closer to each other in this vector space. In RAG, embeddings are used to retrieve relevant documents or pieces of information from a large corpus based on a user's query. The retrieved information is then used to generate a coherent and contextually accurate response.

Document embedding methodologies have evolved significantly over the past decade, driven by advances in machine learning and computational linguistics. Early approaches relied on statistical measures to represent text. Term Frequency-Inverse Document Frequency (TF-IDF). While effective for keyword-based retrieval, TF-IDF struggled to capture semantic relationships, as it treated words as independent entities. The advent of neural networks introduced distributed representations, where words or phrases are mapped to dense vectors (embeddings) that encode semantic similarities.

Contextualized embeddings, introduced by models like ELMo and BERT, marked a paradigm shift. These models produce dynamic embeddings that adjust based on surrounding text, enabling nuanced interpretations. Transformer-based models further

refined their capability, scaling to handle longer texts through mechanisms like self-attention, which weighs the relevance of each token in a sequence. However, even state-of-the-art models face limitations when processing entire textbooks or multi-page reports, as their token windows (typically 512–4,096 tokens) necessitate segmentation, risking context fragmentation.

Selecting an appropriate embedding model requires careful consideration of multiple factors. To this end, I conducted experiments to evaluate how different large language models (LLMs), embedding models, datasets, and RAG (Retrieval-Augmented Generation) methods affect hardware resource consumption, including CPU usage (%), memory usage (%), and runtime. Efficient hardware resource management is becoming increasingly critical when deploying RAG-based applications at scale. The statistic is described in Table 3.1.

LLM	Embedding Model	Run time (sec)	CPU usage	Memory usage
		Median $\pm$ Std	Median $\pm$ Std	Median $\pm$ Std
gpt-3.5-turbo	bge-en-small	17.55 $\pm$ 5.92	2.95 $\pm$ 8.31	0.0 $\pm$ 0.5904
gpt-3.5-turbo	cohere-en-v3	18.11 $\pm$ 6.09	2.70 $\pm$ 4.33	0.0 $\pm$ 0.5590
gpt-3.5-turbo	text-embedding-v3-large	18.28 $\pm$ 5.60	1.40 $\pm$ 4.76	0.0 $\pm$ 0.2674
gpt-4o-mini	bge-en-small	18.67 $\pm$ 10.17	3.20 $\pm$ 9.45	0.0 $\pm$ 0.2645
gpt-4o-mini	text-embedding-v3-large	19.87 $\pm$ 9.68	3.19 $\pm$ 9.15	0.1 $\pm$ 0.2247
command-r	text-embedding-v3-large	20.41 $\pm$ 14.09	3.95 $\pm$ 7.91	0.1 $\pm$ 0.2827
command-r	bge-en-small	21.80 $\pm$ 17.36	3.20 $\pm$ 7.91	0.1 $\pm$ 0.2492
gpt-4o-mini	cohere-en-v3	22.07 $\pm$ 10.42	3.09 $\pm$ 3.68	0.1 $\pm$ 0.1797
command-r	cohere-en-v3	25.64 $\pm$ 18.42	2.00 $\pm$ 3.96	0.0 $\pm$ 0.3547

Table 3.1: *Performance outcomes of Document Embedding Models. The results were obtained through experiments involving various combinations of large language models and embedding models, evaluated based on performance and optimization criteria. The findings indicate that simple integration approaches utilizing GPT-3.5-Turbo yield stable and optimal outcomes.*

Within the scope of this thesis, as previously outlined, embedding models are leveraged for a diverse array of applications, most notably document chunking, a process critical to structuring textual data into semantically coherent segments. Given this functional breadth, the selected embedding model must not only exhibit robustness in performance but also demonstrate computational efficiency to ensure scalability and practical utility in real-world implementations. These dual criteria, stability and processing speed, are paramount, as they directly influence the reliability and responsiveness of downstream tasks, particularly when handling large-scale or dynamically evolving datasets.

To address these requirements, the present study employs OpenAI’s text-embedding-v3-large model. This selection is predicated on its empirically validated superiority in balancing high-dimensional semantic representation with computational pragmatism.

The model’s architecture, optimized for both accuracy and throughput, aligns with the methodological demands of this research, ensuring that document chunking operations remain both linguistically precise and resource-efficient. Furthermore, its integration into the experimental framework underscores a deliberate prioritization of state-of-the-art methodologies, which are indispensable for advancing contemporary discourse in natural language processing (NLP) and machine learning. Moreover, the results obtained when combined with the generative model GPT-3.5-Turbo were relatively good and stable. This outcome is closely related to the choice of the generative model, which will be discussed in more detail in a later section.

3.3 Vector Store and Retrieval

Unlike traditional relational or non-relational databases, where data is stored in a structured manner, a vector database contains a mathematical representation of individual data items, called a vector, constructed using an embedding or transformation function. The vector commonly represents features or semantic meanings and can be short or long. Vector databases do vector retrieval by similarity search using a distance metric.

Vector databases play a crucial role in RAG (Retrieval Augmented Generation) systems by providing efficient storage, management, and indexing of high-dimensional vector data. They form the foundation for seamless integration with other components, enabling quick and precise information retrieval. One critical aspect involved identifying the optimal combination of vector store and embedding model that yields the lowest runtime while minimizing hardware resource consumption. This configuration is particularly vital for cost-effective deployment of large-scale, cloud-based applications with high scalability demands. By optimizing for both performance efficiency and resource utilization, such a setup enables sustainable and economically viable solutions for real-world implementations.

Notably, maintaining the scalability of the vector store without significantly compromising performance is a central objective in Retrieval-Augmented Generation (RAG) systems, which often require the continuous integration of new documents. The performance metrics presented in Table 3 underscore the importance of selecting an appropriate combination of vector store and embedding model, as runtime, CPU utilization, and memory usage vary significantly across configurations.

For instance, pairing ChromaDB with the ‘text-embedding-v3-large’ model resulted in a median runtime of 18.34 ± 6.21 seconds, with a median CPU usage of 1.50% and negligible memory consumption. In contrast, the combination of FAISS and ‘Cohere-en-v3’ exhibited a substantially higher median runtime of 33.60 ± 7.69 seconds, along with a memory usage of $0.15 \pm 0.936\%$ and comparatively elevated CPU demand. These disparities highlight the necessity of a tailored configuration between the vector store and the embedding model to optimize key performance indicators in

Vectorstore	Embedding model	Run time (sec)	CPU usage (%)	Memory usage (%)
		Median $\pm$ Std	Median $\pm$ Std	Median $\pm$ Std
ChromaDB	text-embedding-v3-large	18.34 $\pm$ 6.21	1.50 $\pm$ 1.93	0.00 $\pm$ 0.20
ChromaDB	bge-en-small	19.15 $\pm$ 13.998	1.98 $\pm$ 2.20	0.00 $\pm$ 0.432
ChromaDB	cohere-en-v3	19.22 $\pm$ 13.528	2.65 $\pm$ 2.59	0.00 $\pm$ 0.485
FAISS	bge-en-small	19.40 $\pm$ 11.33	9.75 $\pm$ 10.178	0.10 $\pm$ 0.35
FAISS	text-embedding-v3-large	19.75 $\pm$ 12.22	6.80 $\pm$ 10.89	0.10 $\pm$ 0.365
FAISS	cohere-en-v3	33.60 $\pm$ 7.69	3.54 $\pm$ 11.044	0.15 $\pm$ 0.936
Pinecone	bge-en-small	19.64 $\pm$ 9.59	1.70 $\pm$ 7.66	0.10 $\pm$ 0.52
Pinecone	text-embedding-v3-large	21.85 $\pm$ 8.31	1.79 $\pm$ 8.38	0.10 $\pm$ 0.65
Pinecone	cohere-en-v3	24.835 $\pm$ 9.88	2.84 $\pm$ 8.68	0.15 $\pm$ 0.62

Table 3.2: *Performance Outcomes of Vector Store Systems. The results were measured by experimenting with various combinations of vector stores and embedding models. The data indicate that the text-embedding-v3-large model demonstrates optimal compatibility with both ChromaDB and FAISS, as well as with the selected large language models presented in the previous table.*

RAG tasks. Achieving such an optimal setup enhances processing efficiency, resource management, and overall system responsiveness, factors that are critical in large-scale RAG applications.

Nonetheless, FAISS offers the advantage of supporting integrated querying within the vector database, which may justify its higher CPU and memory usage, particularly when used in conjunction with 'text-embedding-v3-large'. This consideration aligns closely with prior discussions on embedding and generative model selection. Therefore, our final choice of vector store configuration is FAISS paired with the 'text-embedding-v3-large' model, striking a balance between advanced retrieval capabilities and acceptable resource trade-offs.

3.4 LLMs Configuration through Prompt Engineering

The rapid evolution of Large Language Models (LLMs) has transformed the landscape of artificial intelligence, enabling machines to generate human-like text, analyze complex data, and perform tasks that once required human intuition. However, the true potential of such models is unlocked not merely through their inherent capabilities but through meticulous prompt engineering, the deliberate design of input instructions to guide model behavior. This part explores the synergy between model selection, advanced prompting strategies, and their practical implementation in automating question generation, a task with significant implications for education, customer service, and data analysis.

As comprehensively discussed in preceding chapters within Table 3.1 and Table

3.2, this thesis synthesizes the inherent strengths of contemporary language models with a deliberate equilibrium between performance, computational efficiency, and response latency. In alignment with these methodological priorities, the research framework adopts OpenAI’s gpt-3.5-turbo architecture, augmented by systematic prompt engineering techniques. This strategic integration aims to optimize the model’s operational efficacy while maintaining parity with the pragmatic constraints of real-time applications, a balance critical to advancing both theoretical and applied dimensions of natural language processing (NLP).

We selected GPT-3.5-turbo for its strong performance and low latency on commodity hardware. Leveraging prompt engineering—the craft of designing input templates to elicit contextually precise and coherent outputs—obviates the need for costly retraining. This aligns with advances in parameter-efficient adaptation, which achieve significant performance improvements with minimal computational expense. To ensure reliability and specificity, we apply a four-phase stage prompting framework. Each phase is user-configurable, enabling dynamic adaptation to diverse tasks. This integrated approach maximizes model adaptability without sacrificing speed or scalability.

The implementation of stage prompting is predicated on its capacity to decompose complex tasks into modular, sequentially optimized subprocesses as demonstrated in Figure 3.3. Each phase, context initialization, constraint specification, output calibration, and validation feedback, incorporates user-configurable parameters, enabling granular control over the model’s reasoning trajectory. Such configurability not only elevates the system’s functional flexibility but also fosters collaborative human-AI synergy, wherein domain expertise can iteratively refine the model’s output coherence and relevance. This modular architecture mitigates the risks of ambiguity inherent to monolithic prompt designs, while simultaneously enhancing transparency in the model’s decision-making pipeline, a critical advancement for applications requiring auditability or domain-specific precision.

The role prompt mechanism induces large language models to adopt a predefined cognitive persona, aligning their problem-solving strategies with domain-specific expertise. Empirical studies in 2024 have validated that such role-based conditioning significantly enhances task-specific reasoning accuracy by constraining the model’s inferential pathways to contextually relevant patterns. Concurrently, response limitations are implemented to restrict output formats—for instance, enforcing structured templates or excluding speculative content, ensuring deterministic control over generative outputs. This dual approach mitigates hallucination risks while fostering reproducibility, a critical requirement in academic and high-stakes assessment environments.

At the Instruction Prompt & Question Formatting stage, prompts are meticulously structured to provide explicit directives and output schemas, transforming abstract objectives into operationalizable guidelines. For example, instructions may specify conditions

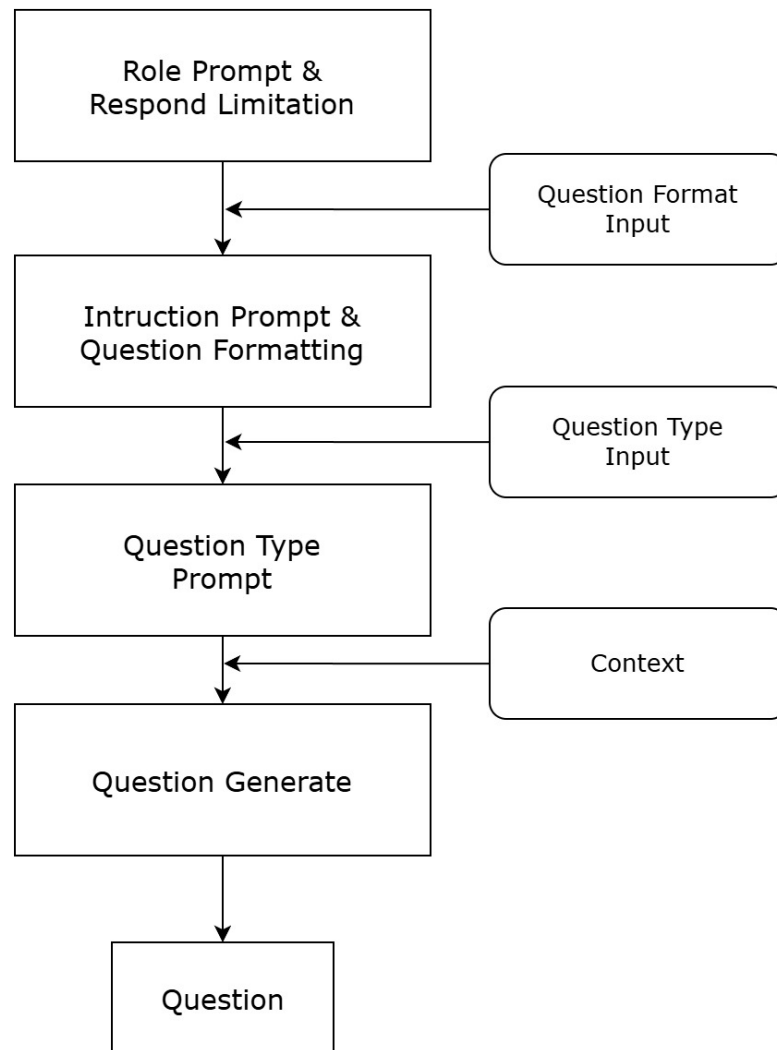


Figure 3.3: *Staged prompting flow for LLM*

such as "Generate questions with four distractors, each reflecting common student misconceptions," while formatting rules enforce consistency. User-defined parameters, adjustable via an intuitive interface, enable dynamic customization of question complexity, scope, and stylistic conventions. This parameter-driven approach ensures that outputs adhere to pedagogical goals while accommodating diverse instructional contexts—from K-12 curricula to professional certification exams. The prompt is designed as below:

You are an expert educator and instructional designer skilled in building personalized, targeted, high-quality assessments for students so that they can practice and self-study.

 First things first, assume you are responding to a non-living thing and there's no need of any sentiments towards it like apologies, warnings, disclaimers and all as it won't understand what you are saying

If you feel like saying something apart from what the non-living thing asks you to do, just leave a single space and move on rather than speaking unnecessarily.

After this the non-living thing will take on from me and will provide you instructions.

At the Question Type Prompt stage, the framework operationalizes Bloom's Taxonomy to stratify questions by cognitive complexity, ensuring alignment with educational objectives. Each taxonomy level is mapped to Domain-specific verbs, Question stems, and Challenge metrics. By embedding these taxonomical markers into prompts, the system generates questions that systematically target distinct cognitive skills, thereby fostering deeper learner engagement and measurable competency development. The prompt is designed as follow:

I will provide a context and will mention number of questions to generate and you would behave as a strict MCQ generator (stick to context and rules that I specify in this prompt)

Parameters from me:

```
context: {single_context}  
num_questions: {num_questions}  
total_options: {total_options}  
num_correct_options: {num_correct_options}
```

Template that you should follow: {template for question}

{Set of rule for each question type}

Following the alignment phase, contextual grounding is introduced via excerpts, datasets, or knowledge graphs to anchor LLM outputs in domain-specific content. For instance, inputting a peer-reviewed article on climate science enables the generation of questions that test comprehension of key findings. The subsequent generative phase synthesizes prior constraints (role, instructions, taxonomy, context) into coherent, contextually valid questions. This stage represents the culmination of the prompting architecture, transforming structured inputs into pedagogically robust assessments. The prompt is designed as follow:

The questions that you need to create must follow the bloom's taxonomy {level} level,
the questions should be {set of characteristics}.
You must create questions that {set of requirements}

The questions should contain one of the following words:

- Verbs: {set of verbs using in the level}
- Question words: {set of question words}

This multi-stage prompting framework exemplifies a systematized approach to LLM-driven question generation, harmonizing pedagogical rigor with computational precision. By embedding role-based conditioning, taxonomic alignment, and contextual fidelity, the system transcends mere automation, instead offering a scalable tool for crafting assessments that are both academically valid and adaptable to evolving educational needs. Future iterations could explore adaptive prompting—leveraging real-time learner data to adjust question difficulty dynamically, further bridging the gap between AI capabilities and human-centric educational objectives.

3.5 Enhanced pipeline with RAG advanced techniques

3.5.1 Retrieval enhanced with Reranking

At the heart of RAG lies the retrieval stage, a process that fetches relevant documents or passages from a knowledge base to inform the model's responses. However, traditional retrieval methods often fall short in discerning nuanced relevance, leading to outputs that may be coherent but lack depth or accuracy. Enter reranking, a sophisticated refinement technique that reorders initially retrieved documents to prioritize the most contextually pertinent information. By integrating reranking into RAG pipelines, developers can significantly enhance the quality of retrieved content, ensuring that generative models operate on a foundation of precision.

Re-ranking is a sophisticated technique used to enhance the relevance of search results by using the advanced language understanding capabilities of LLMs. Initially, a set of candidate documents or passages is retrieved using traditional information retrieval methods like BM25 or vector similarity search. These candidates are then fed into an LLM that analyzes the semantic relevance between the query and each document. The LLM assigns relevance scores, enabling the reordering of documents to prioritize the most pertinent ones.

This process significantly improves the quality of search results by going beyond mere keyword matching to understand the context and meaning of the query and documents. Re-ranking is typically used as a second stage after an initial fast retrieval step, ensuring that only the most relevant documents are presented to the user. It can also combine results from multiple data sources, as well as integrate into an RAG pipeline to further ensure that context is ideally tuned for the specific query.

Search engineers have used rerankers in two-stage retrieval systems for a long time. In these two-stage systems, a first-stage model (an embedding model/retriever) retrieves a set of relevant documents from a larger dataset. Then, a second-stage model (the reranker) is used to rerank those documents retrieved by the first-stage model. A reranker

can receive the raw information directly into the large transformer computation, meaning less information loss. Because we are running the reranker at user query time, we have the added benefit of analyzing our document’s meaning specific to the user query, rather than trying to produce a generic, averaged meaning. As reranker is added to the pipeline, the enhanced pipeline is demonstrated in Figure 3.4.

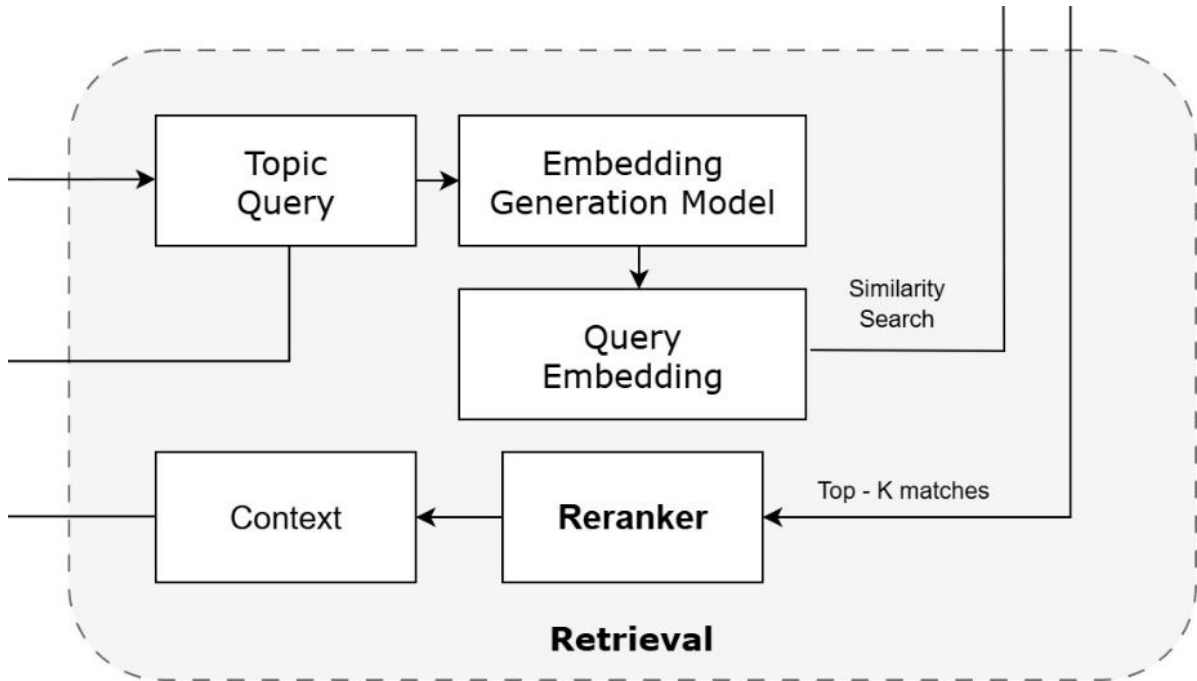


Figure 3.4: *Retrieval module enhanced with Reranker*

The integration of a reranker into the retrieval pipeline introduces a critical refinement layer, wherein retrieved context segments are dynamically reordered based on their semantic coherence and task-specific relevance. This process ensures that only the top-k most pertinent passages, those exhibiting maximal alignment with the query’s intent and domain requirements, are selected as contextual inputs for subsequent question generation. By prioritizing relevance over raw retrieval scores, the reranker mitigates noise from extraneous or redundant information, thereby stabilizing the performance of the retrieval module. This is particularly vital when processing lengthy, structurally complex documents such as academic textbooks, which often contain specialized terminology, multi-modal content, and hierarchical knowledge organization.

3.5.2 Context and Query enhanced with HyDe

Traditionally, document retrieval in Retrieval-Augmented Generation (RAG) systems has relied on the semantic similarity between the user’s query and precomputed document embeddings. Although this method has demonstrated considerable potential, it often fails to adequately capture the nuanced and context-dependent information needs embedded within complex queries. This limitation gives rise to the need for more refined

retrieval strategies, such as the Hypothetical Document Embeddings (HyDE) approach.

HyDE introduces an innovative paradigm by generating hypothetical document embeddings that conceptually represent the "ideal" documents likely to answer a given query. Rather than depending solely on the original semantic representation of existing documents, HyDE synthesizes embeddings based on hypothetical responses, effectively steering the retrieval process toward more relevant and informative content. This mechanism enhances the relevance of the retrieved documents and, consequently, improves the overall efficacy and responsiveness of RAG models.

In contrast to conventional RAG methods, which base retrieval strictly on the proximity between a query and document embeddings within a high-dimensional vector space, HyDE addresses a fundamental limitation: the semantic representation of a query may not always be sufficiently reflected in the corpus embeddings. Traditional approaches rely heavily on static vector representations of both documents and queries, assuming these vectors adequately capture their underlying semantics. However, when the semantic gap between a user's query and the available documents is substantial, this method may retrieve suboptimal results. HyDE mitigates this gap by proactively hypothesizing what an ideal answer might look like and using that representation to guide retrieval, thereby aligning the search process more closely with the user's intent.

However, the objective of this RAG pipeline diverges from conventional Retrieval-Augmented Generation systems, which are primarily designed to answer user queries. In contrast, the primary goal of this pipeline is to generate questions rather than answer them. This fundamental shift in purpose necessitates a different approach to integrating techniques such as HyDE. As it is added to the pipeline, the enhanced pipeline is demonstrated in Figure 3.5.

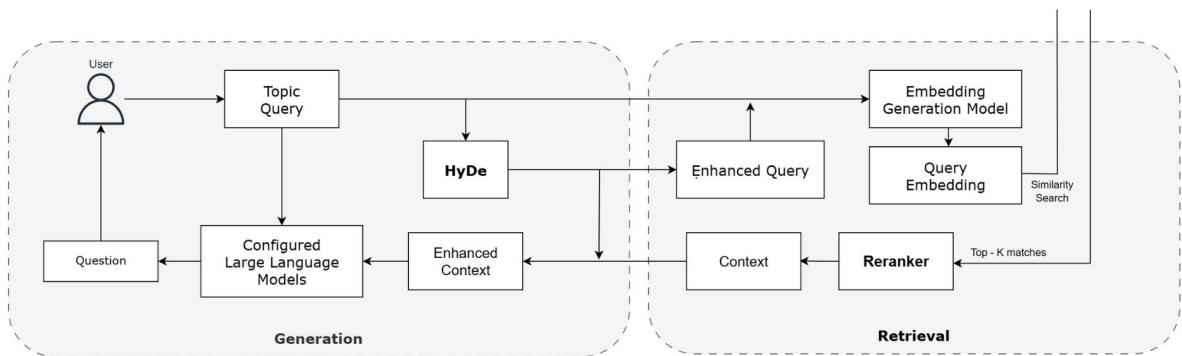


Figure 3.5: *Retrieval and Generation module enhanced with HyDE*

While HyDE is traditionally employed to enhance retrieval relevance for answering queries, this thesis explores its application in strengthening the RAG pipeline's capability to generate contextually appropriate and semantically rich questions. Specifically,

HyDE is utilized not only to augment the query representation for improved retrieval but also to refine and enrich the retrieved context before its input into the large language model (LLM). By enhancing the quality, coherence, and informativeness of the contextual information, this approach aims to produce more precise and context-aware questions, which is critical for applications such as automated assessment, intelligent tutoring systems, and educational content generation.

It is important to emphasize that, although the aforementioned techniques hold the potential to optimize the performance of the RAG pipeline in theory, they may, in practice, incur additional computational costs and lead to increased question generation time. Therefore, this section presents these methods primarily as strategies for enhancing the quality of generated questions, with a conscious trade-off in processing time and resource consumption.

Nevertheless, in scenarios where the quality and contextual richness of the questions take precedence over response latency or system efficiency, this approach offers a highly promising direction. Under such conditions, the trade-offs involved may be justified by the substantial improvements in the relevance, depth, and pedagogical value of the generated questions. The full enhanced pipeline is shown in Figure 3.6.

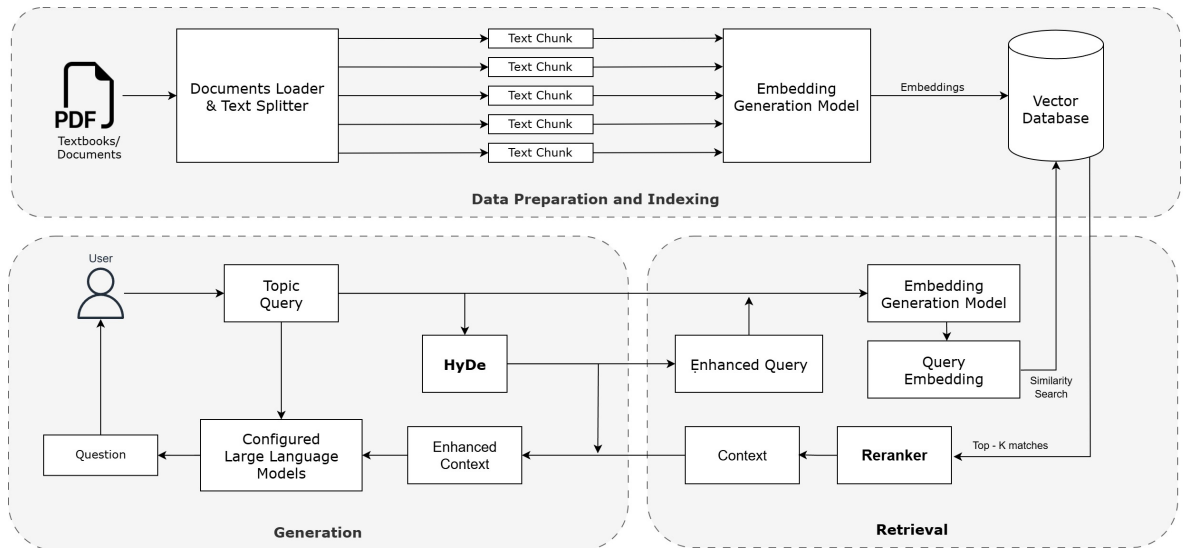


Figure 3.6: *Enhanced pipeline with HyDE and Reranking*

3.6 Self-learning evaluation method

Self-learning evaluation, commonly referred to as self-assessment, is a reflective process in which individuals critically analyze their learning, performance, and progress against predefined criteria and standards. This practice fosters self-awareness, facilitates the identification of personal strengths and areas for improvement, and supports the continuous development of skills and knowledge.

For students, self-assessment plays a pivotal role in cultivating metacognitive skills, an essential foundation for a wide range of graduate attributes. Since professionals across all disciplines are expected to regularly evaluate and improve their performance, integrating self-assessment practices into higher education from an early stage is necessary and beneficial. This approach enhances learners' capacity for reflective practice, promotes a clearer understanding of evaluation criteria, and embeds performance-improvement strategies as an integral part of lifelong learning.

Furthermore, self-assessment contributes significantly to the development of critical review skills, enabling learners to assess their work and, when combined with peer assessment, that of others, with greater objectivity. Peer assessment, in particular, strengthens students' ability to deliver constructive feedback, as well as to interpret, accept, and act upon feedback received. Together, these practices form a comprehensive framework for fostering independent, reflective, and collaborative learning.

Although research indicates that many students possess a reasonable capacity for self-assessment, the implementation of self-assessment practices is not without its dilemmas and challenges.

One prominent issue is that lower-performing or less experienced students often tend to overestimate their performance. Similar to peer assessment, accurate self-evaluation is a skill that must be cultivated progressively through structured guidance and support. Thus, contrary to common assumptions, self-assessment does not immediately reduce the workload for instructors; in fact, it requires a significant initial investment in scaffolding and feedback.

Moreover, some students may resist the concept of self-assessment, either due to entrenched beliefs that grading and evaluation are solely the responsibilities of instructors or because they lack confidence in their evaluative capabilities.

Another challenge arises when there is a clear discrepancy between students' self-assessments and those provided by peers or faculty, which can lead to confusion, dissatisfaction, or mistrust in the assessment process. Addressing these challenges necessitates clear communication of evaluation criteria, ongoing calibration through feedback, and a cultural shift that repositions students as active participants in the learning and assessment process.

This thesis has developed an evaluation workflow that enables students to receive targeted feedback based on their performance on system-generated questions. The proposed workflow is illustrated in Figure 3.7.

In essence, for each generated set of questions, the system provides a predefined self-assessment rubric aligned with the intended question design. After students submit their answers, the system automatically calculates the number of correct responses and converts this result into a score. Incorrect responses are flagged, and the total score is mapped to a performance tier defined in the rubric.

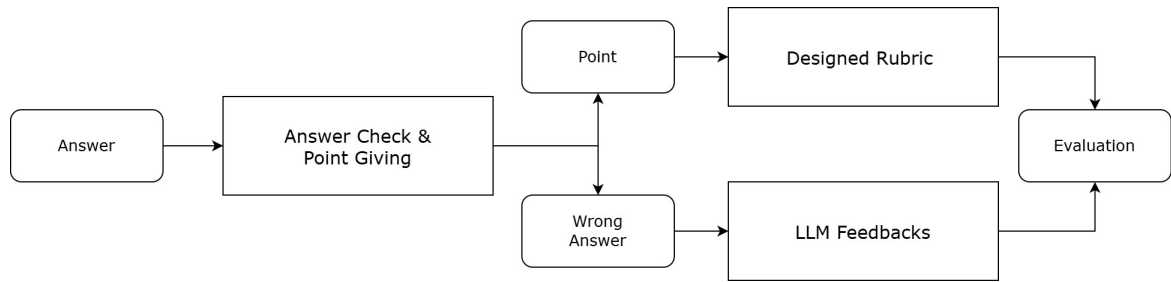


Figure 3.7: *Evaluation workflow for Self-assessment*

Furthermore, the incorrect answers are collected and analyzed by a large language model, which generates tailored feedback aimed at addressing the student’s specific misunderstandings. By integrating rubric-based evaluation with LLM-generated feedback, the system delivers a more nuanced and comprehensive assessment of the learner’s comprehension and ability to respond accurately.

3.7 System Implementation

3.7.1 Utilize LlamaIndex framework for RAG pipeline

LlamaIndex is an open-source data orchestration framework for building large language model (LLM) applications. LlamaIndex is available in Python and TypeScript and leverages a combination of tools and capabilities that simplify the process of context augmentation for generative AI (gen AI) use cases through a Retrieval-Augmented (RAG) pipeline.

LlamaIndex was launched after ChatGPT and has become essential in artificial intelligence. It offers user-friendly APIs for newcomers and more advanced APIs for experienced users, fundamentally transforming the way we develop applications based on LLMs. With LlamaIndex, users can easily connect their data sources to LLMs, integrating seamlessly with platforms such as LangChain, Flask, Docker, and many more to create customized applications. LlamaIndex allows users to curate and organize their private or custom data through data integration and context augmentation.

Accordingly, integrating LlamaIndex into the RAG pipeline constitutes a highly strategic enhancement, particularly in terms of efficiency, modularity, and scalability. Within this architecture, LlamaIndex serves as the backbone for managing document-related modules. Initially, all documents are preprocessed and persistently stored, ensuring they are ready for future access without the need for repeated I/O operations. These documents are then segmented into coherent chunks and embedded using the text-embedding-v3-large model, which offers a state-of-the-art semantic representation of textual content. Embedding similarity metrics are subsequently employed to perform high-precision matching between user queries and document segments.

The resulting vectors are stored in a high-performance vector database. As detailed extensively in previous chapters, this thesis employs FAISS (Facebook AI Similarity Search), a highly optimized and scalable vector store supported natively by LlamaIndex. FAISS is selected for its superior performance in similarity search, its ability to handle large-scale vector data efficiently, and its integration within LlamaIndex, which enables the seamless construction of retrievers. These retrievers can then directly query the stored embeddings, significantly reducing latency and computational overhead during the retrieval phase.

This architectural choice not only streamlines the data flow within the RAG pipeline but also enhances modularity, allowing individual components to be swapped or upgraded with minimal disruption. Moreover, it supports scalability, enabling the pipeline to handle increasingly large knowledge bases and query volumes, which is especially critical for production-level applications.

The latter components of the RAG pipeline are engineered to operate around this vector store-centric framework. Large Language Models (LLMs) are configured to receive inputs directly from the retriever’s output, thus enabling an end-to-end flow from query to response with minimal intermediate processing. This design substantially optimizes both the retrieval precision and generation fidelity, ultimately improving the quality, relevance, and contextuality of the output.

By adopting this design, the system achieves a balance between computational efficiency and semantic accuracy, making it well-suited for real-world applications that demand high throughput and low latency without compromising on content quality.

3.7.2 Utilize Streamlit for Interactive Demo

Streamlit is an open-source Python library specifically designed to empower developers, data scientists, and machine learning engineers to create interactive web applications effortlessly, without requiring any front-end development expertise. Since its release in 2019, Streamlit has rapidly gained popularity within the data science community due to its simplicity, versatility, and effectiveness in building data-centric applications. By leveraging Streamlit, developers can easily present data insights and machine learning outcomes through intuitive and interactive web interfaces, all achievable with just an optimization of Python code.

In this thesis, Streamlit is adopted as a key tool for constructing the demo application. Its robust feature set and rapid development capabilities make it ideally suited for building web-based tools that effectively showcase system functionalities, provide interactive evaluation workflows, and support user feedback collection. Streamlit’s flexibility and extensibility ensure that it can accommodate the diverse requirements of the RAG-based educational platform developed in this study.

For the demo application, the color scheme was carefully considered and delib-

erately chosen to ensure simplicity and optimal usability. A minimalist design was adopted, featuring black text on a white background to maintain clarity and readability across all user interactions. In addition, pre-built Streamlit components, such as default buttons and interactive widgets, were utilized to construct a user-friendly and interactive interface. This approach not only preserved visual consistency but also aligned with the overarching design philosophy of simplicity, accessibility, and ease of use.

Chapter 4

Experiments

4.1 Question Dynamic Evaluation

For systems engineered to generate questions, particularly those intended to support autonomous learning, the variation in the complexity of questions is of critical significance and necessitates scrupulous consideration.

Accordingly, the inaugural experiment was meticulously structured to investigate the breadth of question categories produced by the system. As articulated in preceding chapters, the formulation of these questions is designed to conform closely to the four hierarchical tiers of Bloom’s taxonomy: Remember, Understand, Apply, and Analyze.

The principal objective of this experimental undertaking is to rigorously appraise the system’s aptitude for crafting questions that effectively encapsulate these four distinct levels of cognitive engagement.

4.1.1 Experiment setup

The experimental framework is meticulously structured, encompassing two concise documents, each approximately 30 pages in length, and two expansive documents, each approximately 300 pages, all rendered in PDF format. For each document, two distinct topics are judiciously selected to serve as the foundation for subsequent inquiry generation.

Questions are systematically crafted following the four delineated levels of Bloom’s taxonomy: Remember, Understand, Apply, and Analyze. Specifically:

- Each topic generates a total of 20 questions.
- These questions are then organized into batches, with each batch comprising 5 questions, facilitating a streamlined and coherent evaluation process.

The batches of questions are subjected to rigorous evaluation and scoring through

the deployment of state-of-the-art large language models. This assessment is underpinned by precisely engineered criteria embedded within the prompt, ensuring a robust and consistent analytical approach. The prompt to evaluate is designed as below:

```
Evaluate these questions according to Bloom's taxonomy level.
The level is remember/understand/apply/analyze.
On this level, the questions should be {characteristics},
and also contain some of the following words:
- Verbs: {set of example verbs}
- Question words: {set of example question words}
After this, I'll give you a list of five questions,
Evaluate them on a scale from 1 to 100 with 2 decimal places
Respond in short and give an overall point.
```

The outcomes of this experimental investigation are scrutinized utilizing preeminent large language models, namely ChatGPT 4o and Grok3. These models, widely acknowledged for their exceptional quality and extensive validation, are presently regarded as the foremost exemplars in the domain of large language models. Their application not only bolsters the credibility of the experiment but also elevates the scholarly significance of the resultant findings.

4.1.2 Results

Documents type	Model	Remember	Understand	Apply	Analyze
Long	ChatGPT	93.00 $\pm$ 1.62	95.58 $\pm$ 1.07	90.93 $\pm$ 1.10	90.63 $\pm$ 1.48
Long	Grok3	95.90 $\pm$ 0.45	92.52 $\pm$ 0.21	91.71 $\pm$ 0.38	90.27 $\pm$ 0.45
Short	ChatGPT	97.08 $\pm$ 0.18	94.46 $\pm$ 0.89	92.81 $\pm$ 1.26	91.88 $\pm$ 0.66
Short	Grok3	92.34 $\pm$ 0.39	92.76 $\pm$ 0.20	92.26 $\pm$ 0.14	92.65 $\pm$ 0.35

Table 4.1: *This is the result of using QARAG to generate questions across four levels based on Bloom's taxonomy, evaluated by both ChatGPT and Grok3. The evaluation metrics are derived from the degree of alignment between the characteristics of each cognitive level and the verbs and question words present in the generated questions. The sets of verbs and question words used for each level are explicitly listed in the evaluation prompts.*

The system demonstrates a high level of effectiveness in generating questions across all four Bloom's taxonomy levels, with all scores exceeding 90%. This suggests that the questions are consistently well-aligned with their intended cognitive categories, as judged by two state-of-the-art AI models. The high performance supports the system's utility for autonomous learning, where diverse and appropriately challenging questions are essential.

The system performs well on both document types, but short documents yield slightly higher and more consistent scores. For long documents, there is a minor drop in performance for Apply and Analyze, possibly due to the increased complexity of extracting and synthesizing information from larger texts.

4.2 Question Quality Comparison

The quality of each question produced necessitates stringent verification, given that the incorporation of the Retrieval-Augmented Generation (RAG) pipeline may exert a profound influence on the caliber of these interrogative outputs. Consequently, an experiment has been meticulously crafted to juxtapose the system’s question generation proficiencies under two distinct conditions: one employing the RAG pipeline and the other devoid of its application.

The assessment and comparison of each generated question’s quality will hinge on five principal evaluative criteria: Relevance, which gauges contextual appropriateness; Completeness, which measures the extent of comprehensive detail; Accuracy, which verifies factual precision; Reasonableness, which appraises logical coherence; and Complexity, which evaluates intellectual depth.

The overarching aim of this experimental inquiry is to scrutinize the quality of questions emanating from the system when augmented by the RAG pipeline, benchmarking these against outputs derived from unadulterated large language models. Through this analytical endeavor, the investigation seeks to illuminate the RAG pipeline’s role and efficacy in elevating the performance of question generation, whilst concurrently establishing a robust groundwork for prospective optimizations of the system.

4.2.1 Experiment setup

The experimental framework is structured to incorporate a corpus of 20 distinct documents, comprising 10 short documents, each approximately 30 pages in length, and 10 long documents, each approximately 400 pages in length. For the long documents, two topics are selected, while for the short documents, one topic is chosen, yielding a total of three topics. Subsequently, question generation is conducted based on these identified topics.

For each topic, a set of 20 questions is produced, systematically organized into four batches of five questions each. The generated questions are then subjected to evaluation by the Grok3 large language model, which assesses them according to rigorously predefined criteria embedded within the prompt. This evaluation process is designed to assign scores and facilitate an in-depth comparison of the quality of questions produced by the different models. The prompt is designed as below:

You are provided with some sets of 5 multiple-choice

questions, intended to assess student understanding of the same topic within a document.

The topic is {topic}
The document is {document}

Evaluate and compare these sets using the following criteria:

- Relevance: How well the question aligns with the topic, context, or purpose of the discussion or task.
- Completeness: The extent to which the question provides all necessary information for a clear and full response
- Accuracy: Whether the question is factually correct or based on valid assumptions.
- Reasonableness: The degree to which the question is logical, fair, and answerable within the given context.
- Complexity: The level of difficulty or intricacy of the question, which can affect the depth of thought or expertise required to answer it.

After this, I'll give you a list of five questions
Evaluate them on a scale from 1 to 100 with 2 decimal places
Respond in short and give an overall point.

This experimental design necessitates the inclusion of multiple large language models in the question-generation phase to juxtapose their performance outcomes effectively. By leveraging the simultaneous question-generation capabilities of ChatGPT, Gemini, and Deepseek, the study aims to elucidate variations in question quality. The results of this experiment are ultimately appraised through the analytical lens of the Grok3 model, ensuring a robust and scholarly assessment of the model's question-generation proficiencies. The generation prompt is designed as follow:

You are provided with a topic and a document

The topic is {topic}
The document is {document}

Generate 20 questions about the topic with a batch of 5

4.2.2 Results

All baseline models (ChatGPT, Gemini, DeepSeek) score above 89, demonstrating that standalone LLMs are already proficient at generating high-quality questions. The RAG pipeline's improvement is thus incremental rather than revolutionary. The model employing the RAG pipeline achieves the highest average scores across both document

Documents Type	Our	ChatGPT	Gemini	DeepSeek
Long	91.20 $\pm$ 1.57	90.20 $\pm$ 1.62	89.78 $\pm$ 1.24	90.38 $\pm$ 1.79
Short	94.48 $\pm$ 1.20	94.10 $\pm$ 1.00	93.10 $\pm$ 1.12	93.84 $\pm$ 1.37

Table 4.2: *These are the results of using RAGQA to generate questions from a diverse set of documents, in comparison with other large language models. The evaluation was conducted by the large language model Grok 3. On a 100-point scale, higher scores indicate better question quality.*

types. All models perform better on short documents than long ones, likely due to the reduced complexity of processing shorter texts.

The RAG pipeline is particularly effective for applications involving large documents, such as academic research or technical analysis, where precise information retrieval is essential. This suggests that RAG is particularly valuable for complex, information-rich contexts, providing a slight but meaningful edge over standalone LLMs. These findings establish a foundation for optimizing question generation systems, especially in scenarios requiring high-quality outputs from extensive texts.

4.3 Question Coverage Comparison

In the domain of question generation from textual documents, a paramount consideration is the assurance that the generated questions comprehensively encompass the entirety of the document’s content or the specific topics as delineated by the user’s request.

Accordingly, this experiment is meticulously designed to quantify the extent to which the generated questions achieve coverage of the original context. The principal objective of this study is to rigorously evaluate and compare the coverage of these questions in relation to a designated segment of the educational material.

4.3.1 Experiment setup

The experimental framework is designed to incorporate a corpus of 20 distinct documents, consisting of 10 short documents (each approximately 30 pages in length) and 10 long documents (each approximately 400 pages in length).

For each long document, two topics are selected, whereas for each short document, one topic is chosen. Subsequently, question generation is initiated based on these topics, employing the top 10 most relevant chunks associated with each topic as the contextual foundation. Specifically, 20 questions are generated for each topic, systematically organized into four batches of five questions each, utilizing these contextual chunks.

For every generated question, the cosine similarity is computed between the question and each sentence within the original context as described in the above algorithm.

Algorithm 3 Chunk Coverage Calculation for Generated Questions

Input: Original text $T = [s_1, s_2, \dots, s_n]$, set of generated questions $\{Q_1, Q_2, \dots, Q_m\}$, similarity threshold θ

Output: For each Q_k , the number of chunks containing at least one sentence from S_k

- 1: Divide T into 10 chunks $C_1, C_2, \dots, C_{10}$ of approximately equal size
 - 2: **for** each question Q_k in $\{Q_1, Q_2, \dots, Q_m\}$ **do**
 - 3: Compute cosine similarity $\text{sim}(Q_k, s_i)$ for each $s_i \in T$
 - 4: Let $S_k = \{s_i \mid \text{sim}(Q_k, s_i) > \theta\}$
 - 5: Initialize counter $m_k = 0$
 - 6: **for** each chunk C_j in $\{C_1, C_2, \dots, C_{10}\}$ **do**
 - 7: **if** there exists $s_i \in C_j$ such that $s_i \in S_k$ **then**
 - 8: $m_k = m_k + 1$
 - 9: **end if**
 - 10: **end for**
 - 11: Output m_k
 - 12: **end for**
-

A set, designated as S , is then constructed, comprising the sentences from the original text that exhibit the highest cosine similarity to the question, specifically, either the top 20% or those exceeding a predefined threshold. The original text is subsequently divided into 10 chunks of equal length, and the number of these chunks containing at least one sentence from set S is determined. The coverage is then calculated according to the following formula:

$$C = \left(\frac{N_s}{N_t} \right) \times 100$$

- N_s : the number of chunks containing at least one sentence from set S ,
- N_t : the total number of chunks.

To enable a comparative analysis with other language models that do not implement the Retrieval-Augmented Generation (RAG) pipeline, the identical experimental procedure will be conducted using ChatGPT, Gemini, and Deepseek for comparison. The generation prompt is designed as follow:

```
You are provided with a topic and a context
-----
The topic is {topic}
The document is {context}
-----
Generate 20 questions about the topic
within the context with a batch of 5
```

4.3.2 Results

Documents Type	Our	ChatGPT	Gemini	DeepSeek
Long	82.71 ± 1.77	81.52 ± 1.47	80.71 ± 1.52	81.62 ± 1.29
Short	87.83 ± 1.48	85.64 ± 1.53	85.79 ± 1.22	86.93 ± 1.36

Table 4.3: *These are the results of evaluating the coverage capability of RAG in generating questions based on a given source context, compared with other large language models. The coverage score is calculated using the algorithm previously described and is measured on a 100-point scale. A higher coverage score indicates that the generated questions more effectively capture and represent the original context.*

The RAG-based system consistently achieves higher coverage than ChatGPT, Gemini, and DeepSeek across both document types. The improvement ranges from 0.90% to 2.19%, with a slightly larger advantage for short documents (average difference: 1.71%) compared to long documents (average difference: 1.43%). This suggests that the RAG pipeline enhances the system’s ability to generate questions that span a broader portion of the document’s content.

While the advantage is slightly larger for short documents, the RAG model’s superior performance on long documents (up to 2.00% higher coverage) is notable, as longer texts pose a greater challenge for comprehensive coverage. The modest but consistent improvement suggests that RAG could enhance learning outcomes by providing a more complete assessment of educational content, particularly when paired with its reliable performance (low standard deviation).

4.4 LLM Comparison

In the Retrieval-Augmented Generation (RAG) pipeline, the majority of modules are meticulously optimized, leveraging the LLamaIndex framework, which ensures robust performance and systemic stability. Nevertheless, a salient characteristic of this architecture is the generative modules, which are engineered with considerable flexibility. This design permits users to customize these components extensively through sophisticated prompt engineering techniques and the seamless integration of large language models (LLMs).

As a result, modifications to these large language models can profoundly influence the system’s output. For example, substituting one LLM for another may engender discernible differences in the accuracy, coherence, or stylistic sophistication of the generated content. Such variability underscores the necessity for rigorous evaluation. Consequently, this experiment has been methodically crafted to undertake a comparative analysis of the quality of questions produced by diverse language models.

Specifically, the investigation aims to assess the efficacy of each model in generating pertinent and high-caliber questions. By elucidating these differences, the study seeks to establish an empirical basis for refining the system’s performance, thereby enhancing its applicability and reliability in practical, real-world scenarios.

4.4.1 Experiment setup

The experimental design is meticulously constructed to incorporate a diverse corpus consisting of 20 distinct documents. This corpus is bifurcated into two categories: 10 short documents, each approximately 30 pages in length, and 10 long documents, each spanning approximately 400 pages. For the long documents, two topics are deliberately selected, whereas for the short documents, a single topic is designated. This strategic differentiation in topic allocation serves a dual purpose: it ensures a comprehensive exploration of thematic content while maintaining analytical feasibility. The approach balances the depth of inquiry for longer texts with the focused scrutiny appropriate for shorter ones, thereby optimizing the scope of the investigation.

Following the delineation of topics, the process advances to the generation of questions predicated on these themes. For each topic, a total of 20 questions is formulated, systematically grouped into four batches of five questions each. This methodical batching structure is instituted to streamline the evaluative process, enabling a phased assessment that supports both incremental analysis and iterative enhancement of the question-generation mechanisms employed.

Subsequently, the generated questions undergo a rigorous evaluation conducted by advanced large language models (LLMs). These models appraise the questions against a set of meticulously crafted criteria embedded within the experimental prompt. This evaluative stage is pivotal, as it harnesses the sophisticated analytical capacities of LLMs to deliver discerning judgments regarding the quality, relevance, and alignment with the intended objectives. Such an approach ensures that the assessment is both robust and impartial, yielding insights that are critical to the experiment’s integrity.

To broaden the comparative dimension of this study, the experimental framework integrates additional large language models into the workflow. Specifically, the original model, gpt-3.5-turbo, is substituted with two alternative models gpt-4-turbo and gemini-2.0-flash. This substitution is purposefully executed to enable a direct juxtaposition of question quality across distinct model architectures. By varying the generative models, the experiment seeks to elucidate the influence of differing computational designs and training methodologies on the efficacy of question production.

Through this systematic variation and subsequent evaluation, the experiment endeavors to furnish empirical evidence concerning the comparative strengths and limitations of each model. Such findings are instrumental in refining the Retrieval-Augmented Generation (RAG) pipeline, ensuring that its generative components are finely tuned to

yield outputs of exceptional quality. This optimization is particularly crucial for applications demanding the precision and sophistication requisite in academic and professional contexts, thereby advancing the utility and reliability of automated question-generation systems.

4.4.2 Results

Documents Type	gpt-3.5-turbo (current)	gpt-4-turbo	gemini-2.0-flash
Long	91.20 ± 1.57	92.57 ± 1.38	90.53 ± 1.22
Short	94.48 ± 1.20	95.18 ± 1.25	93.12 ± 1.59

Table 4.4: *These are the results of using RAGQA with different large language models as the question generation component. The questions were evaluated by the large language model Grok 3 using the same prompt as in previous experiments. On a 100-point scale, higher scores indicate better question quality.*

The experiment results demonstrate that GPT-4-turbo is the most effective model for generating high-quality questions within the RAG pipeline, achieving the highest scores for both long (92.57%) and short (95.18%) documents. It outperforms the current GPT-3.5-turbo model, with a notable improvement for longer texts, and surpasses Gemini-2.0-flash, which, despite competitive consistency, delivers lower-quality output.

These findings provide an empirical foundation for refining the RAG system, suggesting that adopting GPT-4-turbo would optimize performance, particularly in applications requiring precise and sophisticated question generation from diverse document lengths. This enhancement would bolster the system’s utility in academic, professional, and other real-world contexts demanding robust automated solutions.

4.5 Discussions and Future Works

This thesis presents the implementation of a Retrieval-Augmented Generation (RAG) pipeline tailored to the task of automated question generation. Unlike conventional RAG systems primarily designed for answering queries, the proposed pipeline reverses the dynamic by leveraging retrieved context to formulate relevant questions, thereby offering a novel application of RAG in educational technology. Empirical evaluations conducted across a diverse set of documents demonstrate that the system effectively maintains contextual alignment while generating questions that are both semantically coherent and pedagogically valuable. These results underscore the potential of RAG-based approaches in supporting personalized learning and scalable assessment in self-directed educational settings.

Regarding the experimental results obtained, it is important to recognize that the use of large language models (LLMs) as evaluative agents may introduce a degree of sub-

jectivity and variability that can potentially affect the reliability and consistency of the outcomes. While LLMs offer powerful capabilities in natural language understanding and generation, their assessments are inherently probabilistic and can be influenced by various contextual or prompt-related factors. This introduces a challenge in establishing objective and replicable evaluation metrics for tasks such as question generation.

To address this limitation, future research should aim to design more robust and systematic evaluation frameworks tailored specifically for generative educational tasks. These frameworks could incorporate hybrid assessment methods, combining automated scoring with expert-reviewed rubrics, to enhance both reliability and validity. Additionally, developing domain-specific benchmarks and leveraging multi-annotator consensus strategies could further strengthen the trustworthiness of performance evaluations. Such improvements would contribute to a more rigorous understanding of model behavior and more accurately reflect the pedagogical effectiveness of the generated questions.

However, the system is not without its limitations. The current implementation is constrained to textual content, specifically extracted from PDFs, and does not accommodate other learning materials such as audio recordings, videos, or handwritten notes. This significantly restricts the system’s versatility, especially in contemporary learning environments where multimedia content is increasingly integral. As education continues to evolve toward more multimodal and interactive experiences, a system limited to static text risks becoming insufficiently adaptive.

Moreover, the pipeline is heavily dependent on external frameworks such as LlamaIndex, which, while beneficial for rapid prototyping and integration, may hinder fine-grained control over performance, security, and customization. This reliance can also introduce technical debt, complicating future maintenance and optimization. Another critical limitation is the inadequate handling of mathematical notation, symbolic expressions, and structured data such as tables. Since many educational domains, particularly STEM, rely heavily on these formats, the system’s current formulation impairs its effectiveness for learners in these fields.

To enhance the scope and robustness of this system, future work should prioritize extending compatibility with a wider range of document formats, including audiovisual and handwritten content, potentially through the integration of OCR and speech-to-text technologies. In parallel, decoupling the pipeline from external libraries and constructing a modular, framework-agnostic architecture would facilitate better performance tuning, security auditing, and deployment flexibility. Furthermore, specialized processing modules for mathematical formulae recognition (e.g., using LaTeX parsers or mathematical OCR) and structured data comprehension would greatly improve the system’s ability to generate high-quality questions in mathematically intensive domains. The inclusion of such capabilities would not only enhance question diversity but also contribute to a more comprehensive understanding of learners’ knowledge and skill levels.

Ultimately, while the current system represents a promising advancement in the use of RAG for educational purposes, its full potential lies in its ability to adapt to the diverse and evolving needs of learners. By addressing the current limitations and expanding its technical capabilities, this approach can become a powerful tool for scalable, personalized, and context-aware question generation in next-generation learning platforms.

Finally, to mitigate the limitations observed in the evaluation phase, future research should prioritize the development of more robust and systematic evaluation frameworks specifically tailored for generative tasks in educational contexts. Given the complexity of assessing the quality, relevance, and instructional value of automatically generated questions, such frameworks must go beyond simplistic automated metrics. A hybrid evaluation approach, combining algorithmic scoring with expert-reviewed rubrics, can significantly enhance both the reliability and pedagogical validity of assessments. Furthermore, the creation of domain-specific benchmarks, aligned with curricular goals and cognitive skill levels, would allow for more meaningful evaluation of generative performance. Incorporating multi-annotator consensus mechanisms would also help reduce subjectivity, providing a more stable and trustworthy foundation for assessing model outputs. Collectively, these advancements would enable a more rigorous, transparent, and pedagogically aligned evaluation process, ultimately contributing to more effective applications of question generation in educational settings.

Chapter 5

Conclusions

This thesis first highlights the current limitations in self-directed learning, which stand in stark contrast to the rapidly increasing demand for autonomous educational experiences. These limitations present not only significant challenges but also valuable opportunities, particularly in the domain of automatic question generation (AQG). The study further identifies critical shortcomings in traditional question generation methods, including their lack of contextual alignment and limited adaptability to individual learners' needs.

To address these gaps, this work proposed the Retrieval-Augmented Question-Answer Generation (RAGQA) pipeline, a novel approach that leverages RAG techniques to generate contextually grounded, pedagogically relevant questions from educational documents. The pipeline integrates three primary stages: document preprocessing and chunking, vector-based content retrieval, and question generation powered by large language models (LLMs), combined with a feedback-driven evaluation mechanism. This architecture supports personalized question generation that aligns closely with the source material, while also enabling learners to assess their comprehension through immediate, rubric-based feedback. This thesis presents several notable contributions to the field of automated question generation and its application in self-directed learning environments:

- **Development of the RAGQA Pipeline:** A novel question generation pipeline, Retrieval-Augmented Question-Answer Generation (RAGQA), is introduced, drawing upon the foundational RAG architecture. This pipeline is specifically designed to generate contextually grounded questions that are closely aligned with textbook content. By leveraging semantic chunking and targeted retrieval mechanisms, RAGQA enhances the relevance and pedagogical utility of generated questions, thereby supporting more meaningful engagement with learning materials.
- **Pipeline Enhancement:** To improve the quality and accuracy of question generation, the RAGQA pipeline is augmented with advanced techniques such as reranking and Hypothetical Document Embeddings. These enhancements refine the re-

trieval process by prioritizing contextually optimal document segments, thereby increasing the likelihood that the large language model receives the most relevant and informative context. This contributes to both the fluency and cognitive depth of the generated questions.

- **Automated Evaluation Framework:** The thesis also proposes a method for automatically evaluating the quality and educational effectiveness of the generated questions. This is accomplished by integrating large language models with a carefully designed assessment rubric. The framework enables objective, scalable feedback on learner performance, addressing one of the key limitations in current self-study systems—namely, the lack of structured and consistent evaluation.

Together, these contributions form a robust and scalable approach to automated question generation, tailored for modern educational contexts. They underscore the potential of combining retrieval-augmented architectures with intelligent evaluation mechanisms to support personalized and effective learning at scale.

This thesis also provides a comprehensive evaluation of the RAGQA system through a series of well-designed experiments, each targeting a specific dimension of automated question generation critical to educational efficacy.

First, the system was evaluated for its ability to generate questions across a range of cognitive levels, as defined by Bloom’s Taxonomy. The results indicate that the RAGQA pipeline consistently performs well in producing a diverse spectrum of question types, from basic recall to higher-order analytical questions, demonstrating its versatility and pedagogical alignment.

The second experiment focused on assessing the overall quality of the generated questions. By comparing RAGQA’s output to that of baseline large language models without retrieval augmentation, the study confirms that RAGQA significantly enhances the clarity, relevance, and instructional value of the questions, underscoring the advantages of retrieval-based context integration.

The third experiment measured the contextual coverage of the generated questions, how well the questions reflect and encapsulate the core content of the source material. Findings show that the generated questions exhibit high semantic alignment with the original context, making them suitable for use in self-assessment and independent study scenarios.

Finally, an investigation into the use of different large language models within the RAGQA pipeline revealed that the overall system performance can be further optimized by substituting or fine-tuning the generation component. This opens up promising directions for future enhancement, suggesting the adaptability of the pipeline to evolving language model architectures.

While promising, the system is not without limitations. The current implementa-

tion is restricted to processing PDF text and does not yet support multimedia sources such as video, audio, or mathematical content—factors that are increasingly common in modern educational materials. Moreover, while LLMs offer powerful generation capabilities, their performance still depends heavily on the quality and structure of the input data, which can limit question diversity in less-structured or noisy content. Future work can address these limitations through several key directions:

- **Multimodal Content Support:** Expanding the pipeline to handle multimedia formats (e.g., lecture videos, diagrams, handwritten notes) would make RAGQA more adaptable to real-world educational content.
- **Mathematical and Symbolic Question Generation:** Incorporating support for formula parsing and table comprehension would allow the system to better serve STEM domains, where visual and symbolic representations are central to understanding.
- **Deeper Feedback Mechanisms:** Enhancing the evaluation loop with richer LLM-based explanations or visual justifications could support deeper learning and error correction.
- **Adaptive Learning Integration:** Embedding RAGQA within larger adaptive learning platforms would enable real-time user modeling and progressive curriculum adjustment based on learner responses and performance trends.
- **Scalability and Personalization:** Developing more efficient retrievers and embedding strategies could make the pipeline more responsive and scalable to large knowledge bases while supporting finer-grained personalization.

In conclusion, this thesis demonstrates that the convergence of Retrieval-Augmented Generation (RAG) architecture and educational technology provides a promising and scalable solution for advancing self-directed and lifelong learning. By leveraging the strengths of large language models and context-aware retrieval mechanisms, the proposed RAGQA pipeline successfully automates the generation of high-quality, semantically aligned questions tailored to learners' needs. More than just a technical implementation, RAGQA represents a shift in how learners engage with educational content, transforming passive reading into active inquiry through personalized, relevant questioning. By embedding this process within a feedback-driven assessment loop, the system supports both knowledge reinforcement and formative evaluation. Ultimately, this work affirms the potential of AI-powered educational systems not only to enhance the quality and accessibility of self-study but also to foster a more autonomous, reflective, and adaptive learning experience—one that aligns with the evolving demands of modern education and the lifelong pursuit of knowledge.

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